Segment to Focus: Guiding Latent Action Models in the Presence of Distractors

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Abstract

Latent action models (LAMs) offer a promising path to pre-training embodied agents on large amounts of action-free video. They infer latent actions between consecutive observations that can later be decoded to ground-truth actions using a small number of labels. However, recent work has shown that this recipe fails in the presence of action-correlated visual distractors common in real-world video, such as dynamic backgrounds, camera shake, or other moving objects. In these scenarios, the standard reconstruction objective drives latent actions to encode exogenous motion instead of agent-controlled dynamics, resulting in policies that underperform when fine-tuned. We observe, however, that endogenous and exogenous factors are typically spatially separated in pixel space: control-relevant change is concentrated on the agent, while distractor motion occurs elsewhere. We exploit this observation by restricting the reconstruction objective to agent pixels, forcing latent actions to explain agent-controlled dynamics rather than exogenous ones. We call this method **MaskLAM**; it obtains the agent mask zero-shot from off-the-shelf segmentation foundation models (e.g., SAM) and requires no architectural changes, auxiliary losses, or action labels during pre-training. Across two continuous-control benchmarks (Distracting Control Suite, Distracting Meta-World), MaskLAM reduces normalized linear-probe MSE by up to $3.51\times$ and improves normalized return by up to $4.97\times$ over LAPO, while narrowing the gap to LAOM-Labels, which relies on ground-truth action supervision.

1 Introduction

The scalability of reinforcement and imitation learning [2, 17] is heavily limited by the need for action-labeled data. Although the internet contains an abundance of unlabeled video demonstrations, leveraging this resource is still an unsolved problem when ground-truth control annotations are missing. Latent Action Models (LAMs) [7, 24] have emerged as a promising way to overcome this bottleneck: by jointly training an inverse and forward dynamics model on observation-only video, they infer a *latent action* between successive frames, and a lightweight decoder maps these to actual controls using only a small set of labeled examples. On simple benchmarks (static backgrounds and a single moving agent) this approach works consistently well.

Training robust LAMs in realistic, visually-complex environments remains difficult. Standard LAMs rely on *global* reconstruction objectives that force the model to encode the entire scene to predict future

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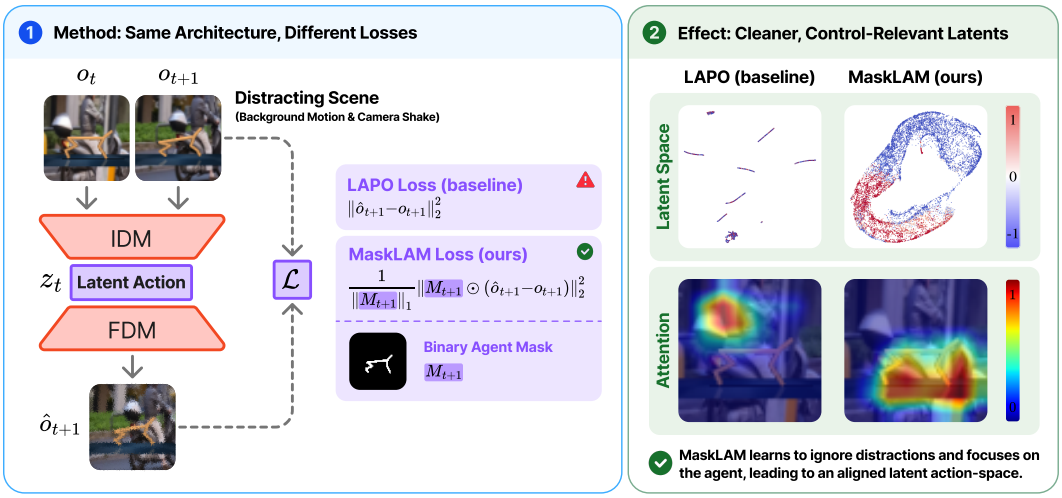


Figure 1: Overview of MaskLAM. Latent action models like LAPO [24] infer a latent action z_t from observation-only video by jointly training an inverse (IDM) and forward dynamics model (FDM). Under visual distractors, LAPO's reconstruction loss pressures z_t to encode exogenous variance and downstream policies fail [20]. Since endogenous and exogenous factors are typically spatially disjoint in pixel space, MaskLAM restricts the loss to agent pixels via a binary mask M_{t+1} obtained zero-shot from SAM 2.1 [23] (1), so distractor pixels contribute no gradient to z_t. (2) On cheetah-run under distractors, the UMAP of the latent action space collapses into entangled clusters for LAPO but forms a smooth, monotonic manifold for MaskLAM (top); Eigen-CAM [19] saliency shifts from the background distractors to the agent (bottom). The intervention adds no auxiliary losses, action labels, or architectural changes to LAPO.

frames. With moving backgrounds, camera shake, or additional moving agents, pixel reconstruction has no reason to prefer agent dynamics over background variation, and the latent encodes whatever varies most [35]. The Exogenous Block MDP (Ex-BMDP) framework [8] formalizes this failure mode: observations entangle a control-endogenous factor (driven by agent actions) with a control-exogenous factor (evolving independently of the agent), and standard reconstruction losses pressure the latent to encode both. As a result, the latent action space of LAPO [24], the basis for many subsequent LAMs, no longer recovers the true action, and downstream policies fail [20]. Prior work addresses this through architectural or modeling interventions: information bottlenecks [33], improved feature extractors [14], auxiliary losses [20, 4], or explicit distractor models [31], at the cost of model complexity, action-label dependence, or extra hyperparameters.

Our key observation is that the Ex-BMDP factorization, although abstract in latent space, is predominantly *spatial* in pixel space: agent actions have immediate, local effect on the *agent's* pixels, while exogenous changes are spatially disjoint from them. Pre-trained segmentation foundation models [23] localize agent pixels zero-shot from a single prompt. Combining the two yields **MaskLAM**: we leave the LAPO recipe unchanged and restrict the forward-dynamics reconstruction loss to agent pixels. Distractor pixels contribute no gradient, so the latent action receives signal only from agent-controlled dynamics. The intervention adds no auxiliary losses, requires no action labels during pre-training, and uses SAM's prompt interface to specify the agent of interest.

Our contributions are:

- We identify spatial separability of agent and distractor pixels as the property that turns the abstract Ex-BMDP factorization into a concrete training-time intervention, and propose **MaskLAM**, a lightweight modification of the LAPO objective that eliminates exogenous gradient on the latent action.
- Across 14 tasks on two benchmarks (Distracting Control Suite, Distracting Meta-World), MaskLAM reduces normalized linear-probe MSE by up to $3.51\times$ and improves normalized return by up to $4.97\times$ over LAPO, narrowing the gap to LAOM-Labels [20], a recent LAM that tackles the same distractor problem by relying on privileged action supervision during pre-training.

- We show that MaskLAM admits compact latent action spaces, lower action-label budgets for downstream decoding, and degrades gracefully under common segmentation failure modes.

2 Related work

Latent action models. Latent action models (LAMs) aim to recover action-relevant representations from unlabeled state transitions. Early work by ILPO [7] learned discrete latent actions through a forward dynamics objective, but mode collapse and poor scalability [28] limited its reach. LAPO [24] resolved both issues by jointly training an inverse and forward dynamics model, and the resulting framework now underpins large-scale robot manipulation pre-training [5, 32] and interactive world models [3]. This progress, however, rests on a fragile assumption: that state transitions are fully explained by agent actions. Once visual distractors enter the scene, this assumption breaks and reconstruction-based LAMs entangle agent dynamics with irrelevant scene variation.

To address this challenge, prior work has focused on architectural and modeling interventions, including improved feature extractors that decompose scenes into object slots [14, 18], auxiliary losses such as action-decoding probes [20] or masked optical flow consistency [4], and explicit distractor models such as AD3 [31] that factorize the world into agent- and distractor-controlled components. While these methods offer improvements, they come at the cost of increased model complexity, action label dependence, or extra hyperparameters, all requiring substantial modifications to the underlying LAM. The closest of them to our work is LAOF [4], which also leverages segmentation masks, but applies them to an auxiliary flow-prediction loss rather than to the LAM's reconstruction objective.

Our approach. MaskLAM keeps the LAM architecture and training pipeline intact, modifying only the reconstruction objective: errors are reweighted by agent-centric segmentation masks, suppressing gradients from distractor regions. Unlike LAOM-Labels, we require no action labels. In contrast to LAOF, the most similar earlier approach that also uses masks, our method applies masks directly to the reconstruction objective rather than on a separate flow-prediction loss. Unlike AD3, it does not rely on any independence assumptions between the agent and distractor dynamics. Unlike object-centric LAMs that mask the inputs, our masks are applied to the loss, thereby preserving the Ex-BMDP formulation (Section 3) and removing the need for slot probing during evaluation.

3 Background and Problem Formulation

Exogenous Block MDP. We adopt the Ex-BMDP framework [8] as the formal foundation for reasoning about distractors. Each observation $o_t \sim Q(\cdot \mid s_t, e_t)$ entangles a *control-endogenous* state s_t, governed by agent actions $a_t \in \mathbb{R}^{d_a}$ via $s_{t+1} \sim T(\cdot \mid s_t, a_t)$, and a *control-exogenous* state e_t, evolving independently via $e_{t+1} \sim T_e(\cdot \mid e_t)$. The goal of latent action learning is to recover a latent action $z_t \in \mathbb{R}^{d_z}$ that captures only the endogenous transition $s_t \to s_{t+1}$, encoding nothing about $e_t \to e_{t+1}$.

Latent action models. Most reinforcement and imitation learning methods require action-labeled trajectories $\tau_n = \{(o_i^n, a_i^n)\}_{i=1}^T$, but the vast majority of video data on the internet provides only observation-only trajectories $\tau_n = \{o_i^n\}_{i=1}^T$. LAMs [7, 24] bridge this gap by inferring latent actions z_i from observation-only data, yielding $\mathcal{D}_z = \{(o_i, z_i)\}$ for downstream policy learning. We build on LAPO [24], the basis for many subsequent LAMs [12, 6, 32]. LAPO jointly trains an *inverse dynamics model* (IDM) $z_t \sim p_{\text{IDM}}(\cdot \mid o_t, o_{t+1})$ and a *forward dynamics model* (FDM) $\hat{o}_{t+1} \sim p_{\text{FDM}}(\cdot \mid o_t, z_t)$ to minimize the next-observation prediction loss:

$$\mathcal{L}_{\text{FDM}}^{\text{LAPO}} = \mathbb{E}\left[\|\hat{o}_{t+1} - o_{t+1}\|_2^2\right]. \tag{1}$$

The latent acts as an information bottleneck: since both models observe o_t but only the IDM sees o_{t+1}, the IDM is forced to compress the difference between consecutive frames into z_t. In distractor-free settings, this difference is most efficiently explained by the agent's true actions [26].

Latent action learning pipeline. The standard pipeline [24] proceeds in three stages:

- *Stage 1* (LAM pre-training): train IDM and FDM on the full unlabeled dataset to learn latent actions.
- *Stage 2* (relabeling and behavior cloning): use the trained IDM to assign latent actions to all transitions, producing $\mathcal{D}_z = \{(o_i, z_i)\}$, and train a latent policy $\pi_{\text{latent}}(z_t \mid o_t)$.

- *Stage 3* (action decoder fine-tuning): using a small set of labeled trajectories, train a decoder $d\colon \mathcal{Z} \to \mathcal{A}$ that maps from latent to ground-truth actions. The deployable policy is $\pi = d \circ \pi_{\text{latent}}\colon \mathcal{O} \to \mathcal{A}$.

Why LAPO fails under distractors. Since $o_{t+1} \sim Q(\cdot \mid s_{t+1}, e_{t+1})$, the FDM in (1) must predict changes from *both* the endogenous and exogenous transitions. Given o_t, z_t is the only varying input to the FDM, so the optimization pressures it to encode exogenous dynamics to reduce prediction error on distractor pixels. Nikulin et al. [20] show that this contamination persists even with a multi-step IDM, larger latents, latent-space FDM, and augmentations, and that ground-truth action supervision during LAM training can bridge the gap; this reintroduces the action-label dependence that LAMs were designed to eliminate.

Problem statement. We seek a method that restores latent action quality in the presence of visual distractors *without* requiring ground-truth action supervision during LAM pre-training, and without modifying the LAM architecture.

4 Method

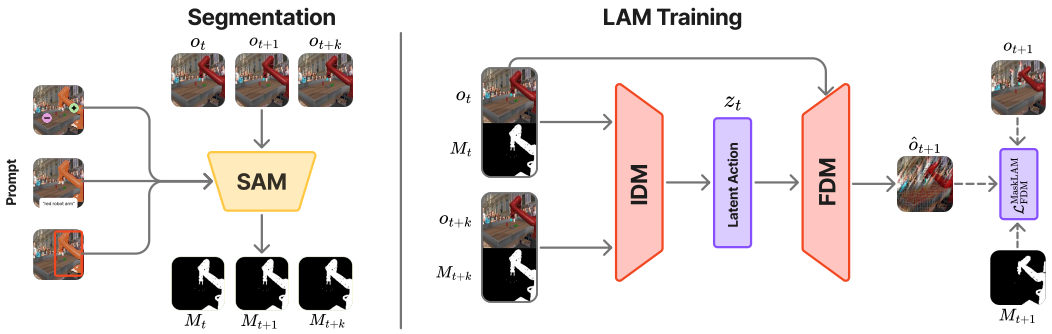


Figure 2: Simplified MaskLAM stage 1 pipeline. Conditioned on a target agent prompt, a pre-trained SAM 2.1 model extracts agent masks M_t, M_{t+1}, M_{t+k} from observations o_t, o_{t+1}, o_{t+k}. The IDM consumes $(o_t, o_{t+k}, M_t, M_{t+k})$ and predicts the latent action z_t; the FDM then reconstructs $\hat{o}_{t+1}$ from (o_t, z_t, M_t). The forward dynamics loss is spatially restricted to agent pixels (white) via M_{t+1}, so distractor pixels (black) contribute no gradient. Endogenous prediction errors thus shape z_t, while exogenous dynamics are filtered out at the loss rather than the encoder. See Section 4.

As shown in Section 3, the FDM loss in (1) reconstructs the full observation, forcing z_t to encode endogenous and exogenous dynamics. Prior work addresses the contamination of z_t at the representation level: action supervision to steer the encoder [20], object-centric slot decomposition [14], or auxiliary optical flow losses [4]. MaskLAM takes a different route. Instead of modifying the encoder or adding auxiliary losses, we modify the *prediction target* to exclude exogenous dynamics from the learning signal. The key observation is that in pixel space, the Ex-BMDP factorization is often *spatial*: endogenous changes correspond to agent pixels, exogenous changes to distractor pixels. Prior methods separate endogenous from exogenous in latent space [8, 15, 14, 4]; MaskLAM performs this separation in pixel space, where pre-trained segmentation models provide a strong, steerable prior.

Spatial separability assumption. Let $M_t \in \{0,1\}^{H \times W}$ be a binary segmentation mask identifying agent pixels, obtained from a pre-trained segmentation model. MaskLAM assumes the Ex-BMDP factorization is approximately spatially separable:

$$M_t \odot o_t \approx M_t \odot Q_s(\cdot \mid s_t), \quad (1 - M_t) \odot o_t \approx (1 - M_t) \odot Q_e(\cdot \mid e_t), \tag{2}$$

where $\odot$ denotes the Hadamard (element-wise) product, and Q_s and Q_e denote the endogenous and exogenous components of the emission function. This holds under a *visual independence* assumption: the rendering of pixels within M_t must be governed mainly by s_t, independent of e_t. Spatial overlap between agent and distractor is allowed.

Masked forward dynamics loss. Given the spatial separability assumption, the most direct intervention is to restrict the FDM loss to agent pixels:

$$\mathcal{L}_{\text{FDM}}^{\text{MaskLAM}} = \frac{1}{\|M_{t+1}\|_1} \|M_{t+1} \odot (\hat{o}_{t+1} - o_{t+1})\|_2^2, \tag{3}$$

where M_{t+1} is the binary mask indicating agent occupancy in the target frame. The loss is zero on distractor pixels, and normalization by $\|M_{t+1}\|_1$ keeps gradient magnitude consistent across varying mask sizes. To see why this eliminates exogenous pressure on z_t, consider the gradient:

$$\frac{\partial \mathcal{L}_{\text{FDM}}^{\text{MaskLAM}}}{\partial z_t} = \frac{2}{\|M_{t+1}\|_1} \left(M_{t+1} \odot (\hat{o}_{t+1} - o_{t+1})\right)^\top \frac{\partial \hat{o}_{t+1}}{\partial z_t}. \tag{4}$$

Where $M_{t+1} = 0$, the residual is zeroed before propagating into z_t. Since encoding $e_t \to e_{t+1}$ cannot reduce $\mathcal{L}_{\text{FDM}}^{\text{MaskLAM}}$, the optimization no longer pressures z_t to capture exogenous dynamics.

Mask-conditioned IDM and FDM. We concatenate the segmentation mask as an extra input channel: the IDM receives $(o_t, o_{t+k}, M_t, M_{t+k})$ and the FDM receives (o_t, z_t, M_t). The mask channel is a *hint*, not a hard gate: the models still receive unmasked observations and can represent the full scene context, including objects the agent interacts with that fall outside M_t (see Figure 5). In principle, conditioning the IDM on the mask also enables multi-agent control, with different masks yielding separate latent action streams from the same observation sequence. We leave this to future work.

Why loss masking, not input masking. Feeding $M_t \odot o_t$ instead of o_t aggressively removes exogenous information but also destroys endogenous context: the endogenous state s_t includes objects the agent interacts with (a gripper approaching a cup, a hand holding a tool); these are action-dependent but may fall outside M_t. Enumerating such objects a priori is impractical, whereas identifying the agent itself is straightforward with pre-trained segmentation. Loss-only masking exploits this asymmetry: the encoder receives full observations and is free to discover the remaining endogenous context from data, while z_t is shaped only by endogenous prediction errors. We evaluate this design choice in Section 6.1.

Implementation (Figure 2)**.** Following prior work [20, 15, 16], we use a multi-step IDM $z_t \sim p_{\text{IDM}}(\cdot \mid o_t, o_{t+k})$ with k sampled uniformly from $\{1, \ldots, 10\}$ during training and $k = 1$ at inference, since exogenous noise decorrelates over longer horizons [15, 8]. As in Nikulin et al. [20], we stack 3 consecutive observations as input to both the IDM and FDM. Masks are obtained zero-shot from SAM 2.1 (hiera-tiny) [23] given a bounding box in the first frame (see Section R), computed in $\sim$10 ms per frame on an A100 and pre-computed for the dataset before LAM training. We use the continuous LAPO baseline from Nikulin et al. [20] (IMPALA ResNet encoder [9], ResNet decoder [11]).

5 Experiment Setup

We evaluate the masked forward dynamics loss (3) along three axes: latent action quality, downstream policy performance, and robustness to common segmentation failures (occlusions, imperfect masks).

Environments. Following prior work on latent actions [20, 14], we evaluate on two benchmarks.

- *Distracting Control Suite* (DCS) [27]: augments DeepMind Control Suite [30] with dynamic natural video backgrounds (DAVIS dataset [21]), continuously shifting camera pose, and randomized agent and object colors. We use the *easy* dynamic setting (magnitude 0.1, 4 background videos, all distractors changing smoothly per timestep). Note that color randomization affects the agent itself, so distractors are not fully spatially separable. We evaluate four continuous-control tasks (cheetah-run, walker-run, hopper-hop, humanoid-walk) at 64×64 pixels. Training/test: 9M/1M transitions.
- *Distracting Meta-World* (DMW): augments Meta-World [34] with dynamic natural video backgrounds (full DAVIS train/val split, 90 videos) at the *easy* dynamic setting. We evaluate the MT10 subset at 128×128 pixels. DMW introduces richer agent geometry (arm, gripper, manipulated objects), testing mask quality under complex interactions. Training/test: 1M/0.1M transitions.

For all environments we provide both ground-truth masks (from simulation) and SAM 2.1 masks [23]; all datasets will be released upon acceptance.

Baselines. *LAPO* [24] trains the IDM-FDM pair without auxiliary losses; following LAOM [20] we remove the VQ-VAE for better performance on continuous control. *LAOM-Labels* adds an auxiliary linear probe loss decoding ground-truth actions during Stage 1 training, establishing an upper bound for what privileged label access can achieve; we use 128k ground-truth labels unless stated otherwise. *LAOF* [4] adds a masked optical flow auxiliary loss, the closest prior method that also leverages segmentation masks, but on flow rather than reconstruction. *LAPO-Slots* [14] decomposes observations into object slots and masks the *input* rather than the loss. We report MaskLAM in two configurations, **MaskLAM@GT** with simulator ground-truth masks and **MaskLAM@SAM** with SAM 2.1 masks [23]; unsuffixed **MaskLAM** refers to both jointly. All baselines use their published hyperparameters, adapted only for convergence on our datasets, with dimension $d_z = 128$ for the latent action space unless stated otherwise; full configurations in Sections B and C.

a) cheetah-run b) window-open

Figure 3: Environments (top: vanilla, bottom: distractor). DCS (a) adds dynamic background videos, agent color randomization, and camera shake; DMW (b) adds dynamic background video distractors only.

Metrics. Three metrics capture different aspects of latent action quality, all normalized against the expert policy that collected the dataset (Section D) and computed over 3 random seeds as mean $\pm$ standard deviation; see Section E for the full protocol.

- *Normalized linear action probe MSE* (NMSE): a linear regressor on frozen latent actions predicts ground-truth actions, with no gradient flowing back [1, 35], normalized by the expert policy's action variance.
- *Normalized success rate* (NSR): mean success fraction over 100 episodes, normalized by the expert collector's rate; reported on DMW following [34].
- *Normalized return* (NR): mean return over 100 episodes, normalized so the expert equals 100%; reported on DCS following [30].

6 Results and Discussion

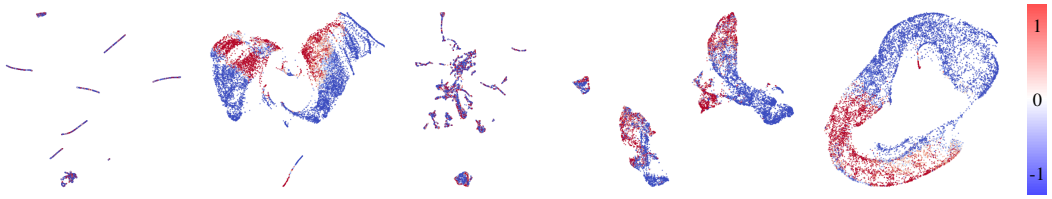


Figure 4: UMAP projection of the latent action space on cheetah-run under the distracting setting of the DCS, colored by the 4th ground-truth action dimension. A well-aligned latent space exhibits a smooth color gradient along the projection. MaskLAM produces a clean, monotonically structured manifold, while LAPO and LAOF collapse into entangled clusters with no clear correspondence to the ground-truth action. Per-dimension projections for all joints are provided in Section P.

We frame our analysis through the lens of Zhang et al. [35], who show that linear latent action models trained by next-frame reconstruction behave like PCA on the change between consecutive frames: the latent encodes whatever varies most across the data, regardless of source. In distractor-free benchmarks the dominant variance is action-driven, and LAPO [24] is able to recover the true actions with almost no additional cost. Once distractors enter the frame, the same loss steers the latent toward

	Distracting Control Suite				Distracting Meta-World			
	Mean NMSE ↓		Mean NR ↑		Mean NMSE ↓		Mean NSR ↑	
Method	**Vanilla**	**Distractor**	**Vanilla**	**Distractor**	**Vanilla**	**Distractor**	**Vanilla**	**Distractor**
LAPO	$0.2971_{\pm 0.1169}$	$0.5114_{\pm 0.0336}$	$0.2364_{\pm 0.0277}$	$0.0759_{\pm 0.0092}$	$\mathbf{0.0742_{\pm 0.0036}}$	$0.3093_{\pm 0.0094}$	$\mathbf{0.8510_{\pm 0.0257}}$	$0.6362_{\pm 0.0147}$
LAOM-Labels	$0.2106_{\pm 0.0060}$	$0.2160_{\pm 0.0034}$	$\mathbf{0.6083_{\pm 0.0165}}$	$\mathbf{0.5663_{\pm 0.0186}}$	$0.1137_{\pm 0.0099}$	$0.1421_{\pm 0.0077}$	$0.7270_{\pm 0.0278}$	$0.7810_{\pm 0.0328}$
LAOF	$0.6390_{\pm 0.0451}$	$0.5056_{\pm 0.0194}$	$0.0721_{\pm 0.0169}$	$0.0939_{\pm 0.0195}$	$0.1849_{\pm 0.0370}$	$0.4529_{\pm 0.0569}$	$0.7160_{\pm 0.1175}$	$0.5768_{\pm 0.0791}$
LAPO-Slots	$0.3516_{\pm 0.0122}$	$0.6570_{\pm 0.0293}$	$0.1264_{\pm 0.0205}$	$0.0454_{\pm 0.0078}$	$0.1090_{\pm 0.0060}$	$0.2450_{\pm 0.0481}$	$0.8326_{\pm 0.0323}$	$0.8274_{\pm 0.0229}$
MaskLAM@GT	$\mathbf{0.2049_{\pm 0.0063}}$	$\mathbf{0.2061_{\pm 0.0093}}$	$0.5126_{\pm 0.0286}$	$0.3776_{\pm 0.0298}$	$0.0815_{\pm 0.0041}$	$\mathbf{0.0880_{\pm 0.0036}}$	$0.8276_{\pm 0.0364}$	$\mathbf{0.8432_{\pm 0.0211}}$
MaskLAM@SAM	$0.2129_{\pm 0.0096}$	$0.2326_{\pm 0.0372}$	$0.4934_{\pm 0.0359}$	$0.3581_{\pm 0.0404}$	$0.0779_{\pm 0.0030}$	$0.0903_{\pm 0.0049}$	$0.8282_{\pm 0.0215}$	$0.8318_{\pm 0.0314}$

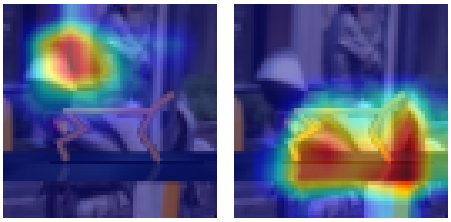
Table 1: Latent action quality (NMSE) and downstream behavior cloning performance (NR/NSR) aggregated over environment suites in the vanilla and distractor settings. MaskLAM@GT and MaskLAM@SAM recover latent actions that align more tightly with ground-truth controls than most baselines, leading to improved downstream performance. This gap increases under distractors.

exogenous variance instead. MaskLAM intervenes at exactly this point: by zeroing the FDM gradient on distractor pixels, it leaves only action-driven variance in the supervisory target. The rest of this section traces the consequences: loss masking restores alignment and downstream performance under distractors, admits smaller latents and label budgets, and remains effective even when the supervisory mask is degraded by occlusion or geometric error.

MaskLAM recovers action-aligned latents under distractors. Across both vanilla and distractor settings, MaskLAM achieves the lowest probe error among label-free methods (LAPO, LAPO-Slots, LAOF), and the gap widens when distractors are added (Table 1). MaskLAM@GT and MaskLAM@SAM are within noise of each other, indicating that off-the-shelf segmentation models are sufficient for MaskLAM to be effective. The aligned latent actions are visible in Figure 4: MaskLAM produces a smooth monotonic gradient over the ground-truth action dimension, while LAPO and LAOF collapse into entangled clusters.

Better alignment translates to better policies. Table 1 reports the corresponding downstream return. MaskLAM@GT matches or exceeds every label-free baseline in both settings, with the largest gap under distractors. MaskLAM@GT also improves over LAPO in the distractor-free setting; we hypothesize that even in clean scenes the unmasked FDM loss diverts a portion of the latent's capacity toward reconstructing background structure rather than action-driven change, an effect we make explicit in the latent-dimension sweep of Figure 6a.

MaskLAM focuses on the agent, not the distractor. To verify that the alignment improvement is driven by the intended mechanism rather than by incidental architectural changes, we visualize the IDM's saliency via Eigen-CAM [19] (Figure 5). LAPO's saliency falls on the distractor background, while MaskLAM's covers the cheetah body and extends to its contact surface with the ground. Part of the saliency falls outside M_{t+1}, indicating that the IDM learns from data, beyond what the mask directly supervises, to attend to the body and its contact with the ground. Full time series in Section O.

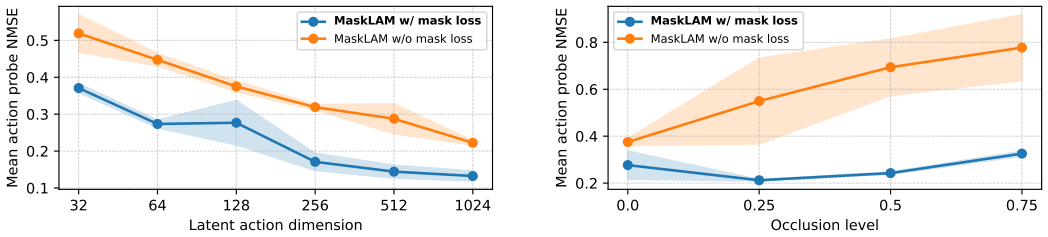


(a) LAPO (b) MaskLAM

Figure 5: Eigen-CAM saliency on cheetah-run DCS (distractor). Warmer colors indicate higher influence on the predicted latent action.

MaskLAM enables compact latent action spaces. Figure 6a sweeps the latent action dimension with and without the masked loss. With masking, a 64-dim latent already matches the alignment of a 256-dim latent without it; the gap is largest in the low-dimensional regime, where encoding extra exogenous content would push out true action information. Nikulin et al. [20] report the same trade-off in LAOM, and Ye et al. [32] observe it in LAPA. By removing the gradient pressure that ties the latent size to the prediction target, MaskLAM lets the model use a small latent without sacrificing alignment.

Cleaner latents allow smaller label budgets. Figure 7 reports downstream return as a function of the number of action labels available for Stage 3 decoder fine-tuning. On DMW (distractor), MaskLAM reaches near peak performance with as few as 2k labels, while LAPO and LAPO-Slots require nearly two orders of magnitude more labels to close the gap. On DCS (distractor) the asymmetry sharpens: LAPO and LAPO-Slots fail to recover a useful policy at any label budget we tested. MaskLAM is strong under limited labels but LAOM-Labels overtakes it at larger label budgets. Since MaskLAM



(a) Latent action dimension (b) Agent occlusion level

Figure 6: Action probe NMSE on DCS (distractor) with and without the mask loss. (a) Latent action dimension (log scale): masking lets a 64-dim latent match a 256-dim unmasked one, with the largest gap in the low-dimensional regime. (b) Agent occlusion level: error stays nearly flat up to 75% occlusion with masking, but more than doubles without it.

actually has lower probe NMSE (Table 1), the gap is not a decoding limit; we hypothesize that direct action supervision yields a latent that holds up better under rollout-induced distribution shift, visible on the longer rollouts on DCS.

MaskLAM degrades gracefully under agent occlusion. Real videos rarely show the full agent at all times. We simulate this on DCS (distractor) by masking out a growing fraction of the agent at training time and measuring the resulting probe error (Figure 6b). With the masked loss, probe error stays nearly flat up to 75% occlusion; without it, the same level of occlusion more than doubles it. Occlusion erodes the target's action-to-noise ratio under the unmasked loss, whereas MaskLAM holds it steady by keeping distractor variance out of the target. Further details in Section K

MaskLAM is robust to imperfect segmentation masks. In practice the supervisory mask comes from an off-the-shelf segmenter, whose boundaries are never pixel-perfect. We perturb ground-truth masks by morphological shrinkage or expansion of 0 to 3 pixels on DWM (distractor), dropping mIoU as low as 0.4, and re-train MaskLAM@GT (Figure 9). Probe error stays below LAPO, LAPO-Slots, and LAOF across the entire perturbation range. Together with the within-noise agreement between MaskLAM@GT and MaskLAM@SAM throughout this section, the takeaway is practical: pixel-perfect segmentation is not required. See mask visualization in Figure 28 for further details.

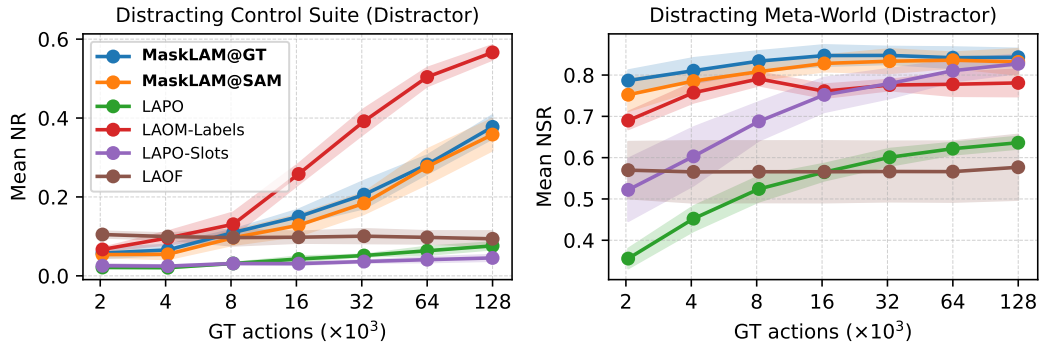


Figure 7: Sample efficiency as a function of the number of ground-truth action labels (log scale) available for action decoder fine-tuning. MaskLAM@GT peaks with as few as 2k labels on DMW (distractor), while LAPO and LAPO-Slots require nearly two orders of magnitude more labels to close the gap, and on DCS (distractor) they fail to recover useful policies at any label budget.

6.1 Ablation Study

To isolate the contribution of each component, we ablate the masked loss (*w/o loss masking*), the additional mask input channel (*w/o mask channel*), both together (*w/o all masking*), and the multi-step IDM (*w/* k=1). We further compare against a variant that zeroes distractor pixels in the input rather

than in the loss (*w/ input masking*). Removing every component reduces MaskLAM to LAPO. Figure 8 reports the resulting action probe NMSE aggregated across DCS and DMW (distractor).

Loss masking is the principal contributor. Removing the masked loss alone more than doubles the alignment error, while turning every component off recovers LAPO. The four perturbations roughly compose back to the LAPO baseline, so the gain over LAPO is decomposable. Loss masking accounts for the largest individual share, consistent with the analysis of Zhang et al. [35]: it is the only component that removes exogenous variance from the supervisory target itself.

The mask channel contributes only in combination with the masked loss. Dropping the mask input channel hurts performance when the masked loss is present, but leaves performance largely unchanged when it is not. The channel acts as a multiplier on loss masking rather than as an independent fix.

Multi-step IDM is a smaller, partially independent gain. Setting $k{=}1$ degrades performance both in isolation and on top of removing all masking. The effect persists with masking turned on, so multi-step IDM is orthogonal to our masking intervention.

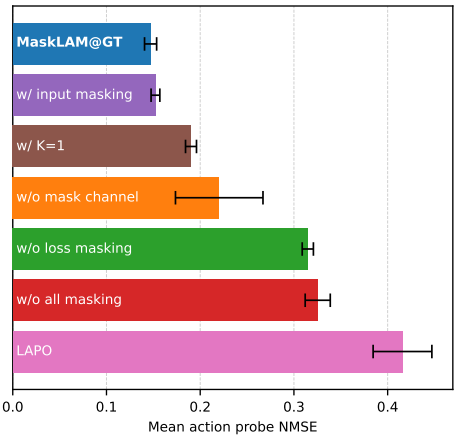


Figure 8: Ablation of MaskLAM components aggregated on DCS and DMW (distractor). Loss masking dominates; the mask channel helps only alongside it; multi-step IDM adds a smaller independent gain.

Input masking matches loss masking quantitatively, not qualitatively. On DCS and DMW (distractor), replacing the masked loss with input masking is statistically indistinguishable from MaskLAM@GT. The mechanism is not: Eigen-CAM (Figure 5) shows the IDM's saliency extends to ground-contact pixels outside M_{t+1}, which loss masking keeps visible but input masking would zero at the input. Parity on these benchmarks reflects that the agent's motion is essentially self-driven, so the contact surface adds little predictive signal beyond the agent pixels. In real video the agent's motion typically also depends on external factors, e.g., a rider on an electric scooter or a commuter on an escalator. There, the interaction surface carries action-relevant signal that input masking discards.

7 Limitations

MaskLAM improves robustness in visually complex settings but has a few practical limitations. Firstly, MaskLAM depends critically on the upstream segmenter: segmentation errors directly corrupt the learning signal for z_t. SAM 2.1 performs well zero-shot but has known failure modes such as prompt ambiguity, out-of-distribution embodiments, and identity drift. In practice the method tolerates substantial mask degradation (Figure 9), so pixel-perfect segmentation is not required;

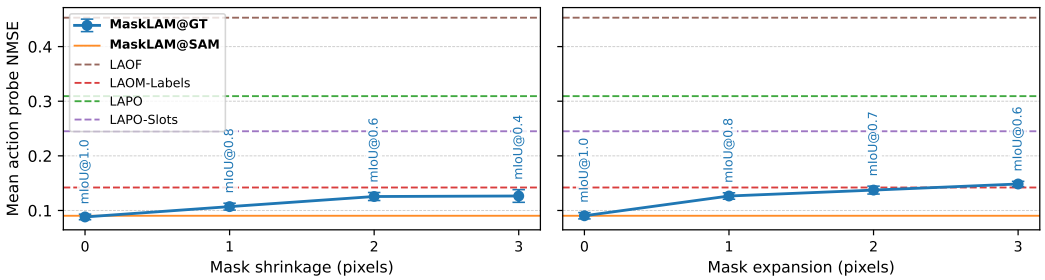


Figure 9: Robustness to imperfect masks on DMW (distractor). MaskLAM@GT is trained with ground-truth masks perturbed by morphological shrinkage (left) or expansion (right) of 0–3 pixels, which drives the mask mIoU down to 0.4–0.6. Baselines are shown as dashed horizontal references. MaskLAM@GT remains below LAPO, LAPO-Slots, and LAOF across the entire perturbation range, indicating that the method **requires no** perfect segmentation to recover well-aligned latent actions.

this is especially true for real-world targets, which lie inside SAM's training distribution unlike simulator environments such as MuJoCo [29]. Secondly, this method inherits limitations of pixel-level reconstruction, including sensitivity to high-frequency noise and increased computational cost and influence of confounding factors like textures on the agent's body. Finally, our evaluation relies on synthetic distractor benchmarks. While they may not capture the full complexity of real-world visual noise, they are widely used and provide controlled, reproducible settings that enable precise analysis of robustness to specific distractor types.

8 Conclusion

In this work, we revisited the problem of learning latent actions from action-free video in the presence of action-correlated distractors. We showed that LAPO's failure in this regime is driven by its prediction target rather than its architecture: as soon as exogenous variance enters the next-frame reconstruction loss, the latent action absorbs it. We proposed MaskLAM, a lightweight modification of LAPO that restricts the forward-dynamics loss to agent pixels using zero-shot masks from an off-the-shelf segmentation model, requiring no auxiliary losses, no architectural changes, and no action labels during pre-training. We found that this single intervention closes the gap to LAOM-Labels, which depends on privileged action supervision. Compared to label-free baselines, MaskLAM additionally yields more compact latent action spaces, lower action-label budgets for downstream decoding, and graceful degradation under agent occlusion and imperfect masks. Our findings suggest that the supervision needed to disentangle agent from distractor dynamics can be re-allocated from action labels, which are scarce, to segmentation masks, which are cheap and already available zero-shot, making latent action pre-training viable on noisy, distractor-rich real-world video.

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A Ethical Considerations and Broader Impacts

MaskLAM lowers the cost of extracting control-relevant signal from action-free video by replacing privileged action supervision with zero-shot segmentation. We summarize the most likely positive and negative consequences of this shift and the segmenter-level failure modes that condition both.

Positive impact. The principal benefit is a reduction in the action-label budget required to pre-train embodied agents on observation-only video. Action labels are scarce and expensive to collect; segmentation masks are cheap and already available zero-shot from foundation models such as SAM 2.1 [23]. By re-allocating the supervision burden from action labels to masks, MaskLAM brings large-scale latent action pre-training closer to the data scales that have driven progress in vision and language, with concrete downstream applications in robot learning, assistive robotics, and any embodied domain where teleoperated demonstrations are difficult to obtain. The same shift broadens the pool of researchers who can experiment with latent action models without privileged hardware or large labeled datasets.

Negative impact. The same property is generic to any video source: a method that recovers control-aligned representations from passively observed video lowers the barrier to building behavioral models of agents that did not consent to that observation. The downstream risks are surveillance-adjacent (an actor with access to public video of a target task can pre-train a policy that imitates the demonstrator without ever instrumenting them, and SAM 2.1's prompt interface makes it straightforward to single out specific individuals or assets within a scene) and automation-displacement-adjacent (any reduction in the cost of imitation from video lowers the marginal cost of replacing the imitated work). We see no technical mitigation at the level of the LAM itself; mitigation belongs at the data and deployment layer, through dataset auditing, restrictions on identifying segmentation prompts, and venue-level norms around consent for video used in embodied pre-training.

B Implementation Details

Observation and stack. Observations are $64\times64\times3$ RGB on DCS and $128\times128\times3$ on DMW. Following Nikulin et al. [20] we stack 3 consecutive frames as input to every module, and the IDM additionally consumes 3 future frames at offset $k \in \{1, \dots, 10\}$ sampled uniformly per training step (Section 4).

Preprocessing. Pixel observations are centered to $[-\frac{1}{2}, \frac{1}{2}]$ via $o \leftarrow o/255 - \frac{1}{2}$, matching the FDM output range. Segmentation masks are binarized to $\{0, 1\}$ at a 0.5 threshold so that ground-truth and SAM masks share the same numerical scale. Ground-truth actions are clamped to the environment action space $[-1, 1]^{d_a}$ before being used as supervision, since the simulator already saturates commands at execution time (Section E).

Encoder backbone. Every convolutional module uses the IMPALA backbone of Espeholt et al. [9]: three encoder blocks with base widths $[16, 32, 32]$ scaled by a channel multiplier m, and 2 residual blocks per encoder block. Each encoder block applies a 3×3 convolution, a 3×3 max-pool of stride 2, and the residual stack, halving spatial resolution; a final flatten and linear projection produces a 128-dimensional embedding. Weights are initialized orthogonally.

IDM. Inputs are the 3 current and 3 future frames concatenated along the channel dimension, with each frame augmented by its segmentation mask as an extra input channel (24 input channels in total at $H\times W$). The IMPALA backbone with multiplier m=6 maps this stack to the continuous latent action $z_t \in \mathbb{R}^{d_z}$ that conditions the FDM. The latent action dimension is $d_z = 128$, if not stated otherwise. No quantizer is used between the encoder and the FDM.

FDM. Implemented as the LAOM-Labels world model of Nikulin et al. [20]: an IMPALA-style encoder with widths $[16, 32, 32]$, multiplier 6, and 2 residual blocks per block, applied to the 3 current frames stacked with their masks (12 input channels). The encoder bottleneck is concatenated channel-wise with z_t projected through a linear layer to the bottleneck spatial shape; a symmetric IMPALA decoder of transposed-convolution upsampling blocks followed by residual blocks reconstructs back to input resolution, and a final 1×1 convolution maps the result to the 3 RGB channels of $\hat{o}_{t+1}$, squashed to $[-\frac{1}{2}, \frac{1}{2}]$ by $\tanh/2$.

Stage 2 latent policy. A fresh IMPALA backbone with multiplier $m{=}4$ takes the 3 current frames only as input (9 channels; no mask, no future frames) and is trained to regress the frozen IDM output via mean-squared error, producing a d_z-dimensional latent action.

Stage 3 BC actor. The Stage 2 latent-policy backbone is loaded and frozen, and a two-layer MLP of hidden size 256 with ReLU activations followed by a linear action head is appended on top of its d_z-dimensional embedding. Only the MLP and the action head are trained, with mean-squared error against the ground-truth action. The actor is therefore fully deterministic at inference.

Baselines. For LAPO, LAOM-Labels, LAOF, and LAPO-Slots we adopt each method's published latent-action backbone unchanged. To keep the comparison fair we set the latent action dimension to $d_z = 128$ across all methods and use the shared Stage 2 latent policy and Stage 3 BC actor described above for behavior cloning and action decoding. Every reported policy is therefore fully deterministic at inference, and any gap in downstream return reflects the quality of the upstream latent action rather than differences in policy architecture.

Training pipelines. LAPO [24], MaskLAM, LAOM-Labels [20], and LAOF [4] all follow the standard three-stage latent-action-learning pipeline of Section 3: Stage 1 trains the LAM on observation-only data, Stage 2 distills a latent policy via behavior cloning against the frozen IDM output, and Stage 3 fine-tunes a small action decoder on a labeled subset. The four methods differ only in their Stage 1 design choices, for which we refer the reader to the original papers; per-stage hyperparameters are listed in Section C.

LAPO-Slots pipeline. LAPO-Slots [14] expands the standard pipeline to five stages by inserting an object-centric pre-training step before LAM training:

- *Stage 1* (slot pre-training): a VideoSAUR encoder is pre-trained on the unlabeled video corpus to convert each frame into a set of object-centric slot representations that separate controllable objects from background distractors.
- *Stage 2* (slot probing): on a small labeled subset, a linear probe regresses ground-truth actions from each slot, and the most action-predictive slot per task is retained.
- *Stage 3* (LAM pre-training): a LAM is trained on the selected slots, inferring latent actions z_t from transitions and learning a forward dynamics model in slot space.
- *Stage 4* (relabeling and behavior cloning): the inferred latent actions serve as pseudo-labels for a behavior-cloning policy that predicts z_t from observations.
- *Stage 5* (action decoder fine-tuning): a small labeled set fine-tunes a decoder that maps the latent policy's output to executable actions.

Hardware. All experiments run on a single NVIDIA A100 40 GB GPU. Peak memory stays below 24 GB across every method and stage, so the pipeline reproduces on commodity consumer cards. We use bfloat16 mixed precision and `torch.compile` where applicable.

Wall-clock per training run. Table 2 reports the total wall-clock for a single end-to-end training run, summed across all stages of the pipeline, for one seed on one task of each benchmark.

Table 2: Wall-clock per end-to-end training run on a single A100 40 GB GPU, summed across all stages of the pipeline for one seed on one task. LAPO-Slots is the most expensive because of the additional VideoSAUR pre-training stage; MaskLAM is the cheapest non-LAPO method.

Method	DCS	DMW
LAPO	$\approx$ 6 h 21 m	$\approx$ 5 h 20 m
LAOM-Labels	$\approx$ 23 h 6 m	$\approx$ 22 h 0 m
LAOF	$\approx$ 11 h 10 m	$\approx$ 7 h 7 m
LAPO-Slots	$\approx$ 37 h 31 m	$\approx$ 18 h 41 m
MaskLAM	$\approx$ 11 h 8 m	$\approx$ 7 h 8 m

Total compute budget. Table 3 reports the total GPU-hours used in this work, aggregated over every seed, task, ablation, and sweep that contributed to the reported numbers. The MaskLAM row is the largest single contribution because of the additional latent-dimension sweep (Section L), the occlusion sweep (Section K), the mask-mIoU sweep (Section M), and the ablation study (Section N).

SAM 2.1 mask pre-computation is accounted for separately as a one-time cost amortised across every MaskLAM@SAM evaluation; see Section R for the underlying throughput numbers.

Table 3: Total GPU-hours used in this work on a single A100 40 GB GPU, aggregated across every seed, task, ablation, and sweep that contributed to the reported numbers. SAM 2.1 mask generation is reported as a separate one-time pre-computation cost.

Method	Runs	GPU-hours
LAPO	959	1,128
LAOM-Labels	588	3,862
LAOF	745	1,007
LAPO-Slots	672	2,566
MaskLAM	2,647	6,843
SAM 2.1 mask generation	—	309
Total	5,615	15,715

Code and data release. We release the full training and evaluation code and the synchronised distractor-free / distracting datasets with both ground-truth and SAM-extracted masks for both benchmarks.

C Hyperparameters

Tables 4 to 8 list per-stage hyperparameters for all five methods. We initialized each baseline from its published hyperparameters and adjusted learning rates, batch sizes, and training budgets where necessary to ensure convergence on our datasets. Pipeline descriptions are in Section B. Where two values appear separated by a slash, the first applies to DCS and the second to DMW; otherwise the value is shared. A dash (—) indicates the hyperparameter does not apply to that stage.

Table 4: Per-stage hyperparameters for MaskLAM (see Section B for the pipeline description).

	Hyperparameter	**Stage 1: LAM**	**Stage 2: BC**	**Stage 3: Decoder**
Optimization	Optimizer	AdamW	AdamW	AdamW
	Learning rate	6×10^{-4} / 3×10^{-4}	4×10^{-4} / 2×10^{-4}	1×10^{-3}
	LR schedule	Cosine annealing	Cosine annealing	Cosine annealing
	Batch size	512 / 128	512 / 128	512 / 128
	Gradient clipping	—	—	1.0
	Mixed precision	bf16	bf16	bf16
Training budget	Epochs / steps	2 / 4 epochs	2 / 4 epochs	25,000 steps
	Block size	13	13	3
	Dataset subset	full	full	128,000
Architecture	Visual encoder	IMPALA	IMPALA	IMPALA
	Channel multiplier	4	4	4
	Encoder channels	6	4	4
	Encoder res blocks	2	2	—
	Latent action dimension d_z	128	128	128
	Frame stack	3	3	3
	Hidden dim (actor)	—	—	256
MaskLAM-specific	Future obs. sampling	✓	×	—
	Future obs. offset	10	10	—
Loss weights	Obs. reconstruction MSE	1.0	—	—
	Action probe MSE	0.01	0.01	—
	Lat. behavior cloning MSE	—	1.0	—
	Behavior cloning MSE	—	—	1.0

Table 5: Per-stage hyperparameters for LAPO [24] (see Section B for the pipeline description).

	Hyperparameter	Stage 1: LAM	Stage 2: BC	Stage 3: Decoder
Optimization	Optimizer	AdamW	AdamW	AdamW
	Learning rate	6×10^{-4} / 3×10^{-4}	4×10^{-4} / 2×10^{-4}	1×10^{-3}
	LR schedule	Cosine annealing	Cosine annealing	Cosine annealing
	Batch size	512 / 128	512 / 128	512 / 128
	Gradient clipping	—	—	1.0
	Mixed precision	bf16	bf16	bf16
Training budget	Epochs / steps	2 / 4 epochs	2 / 4 epochs	25,000 steps
	Block size	4	4	3
	Dataset subset	full	full	128,000
Architecture	Visual encoder	IMPALA	IMPALA	IMPALA
	Channel multiplier	4	4	4
	Encoder channels	6	4	4
	Encoder res blocks	2	2	—
	Latent action dimension d_z	128	128	128
	Hidden dim (actor)	—	—	256
LAPO-specific	Number of latents	4	—	—
	Policy block size	—	3	—
	Future obs. offset	1	—	—
Loss weights	Obs. reconstruction MSE	1.0	—	—
	Action probe MSE	0.01	0.01	—
	Lat. behavior cloning MSE	—	1.0	—
	Behavior cloning MSE	—	—	1.0

Table 6: Per-stage hyperparameters for LAOM-Labels [20] (see Section B for the pipeline description).

	Hyperparameter	Stage 1: LAM	Stage 2: BC	Stage 3: Decoder
Optimization	Optimizer	AdamW	AdamW	AdamW
	Learning rate	6×10^{-4} / 3×10^{-4}	3×10^{-4} / 2×10^{-4}	1×10^{-3}
	LR schedule	Piecewise linear	Piecewise linear	Piecewise linear
	Batch size	512 / 128	512 / 128	512 / 128
	Warmup (epochs)	1	0	0
	Weight decay	0	0	0
Training budget	Epochs / steps	3 / 5 epochs	2 / 5 epochs	20,000 steps
	Frame stack	3	3	—
	Image augmentation	✓	—	—
Architecture	Visual encoder	IMPALA	IMPALA	IMPALA
	Encoder scale	6	32	—
	Encoder res blocks	2	2	—
	Hidden dim (MLP)	—	—	256
	Latent action dimension d_z	128	—	—
	Obs. head dim	1024	—	—
LAOM-Labels-specific	Labeled subset size	128,000	—	128,000
	Labeled batch size	512 / 128	—	—
	Labeled loss coef.	1×10^{-3}	—	—
	Future obs. sampling	✓	×	—
	Future obs. offset	10	—	—
	EMA target τ	1×10^{-3}	—	—

Table 7: Per-stage hyperparameters for LAOF [4] (see Section B for the pipeline description).

	Hyperparameter	Stage 1: LAM	Stage 2: BC	Stage 3: Decoder
Optimization	Optimizer	AdamW	AdamW	AdamW
	Learning rate	3×10^{-4} / 1.5×10^{-4}	2×10^{-4} / 1×10^{-4}	1×10^{-3}
	LR schedule	Piecewise linear	Piecewise linear	Piecewise linear
	Batch size	512 / 128	512 / 128	512 / 128
	Gradient clipping	2.0	—	2.0
	Mixed precision	bf16	bf16	bf16
Training budget	Epochs / steps	2 / 4 epochs	2 / 4 epochs	25,000 steps
	Dataset subset	128,000	128,000	128,000
	Frame stack	—	3	3
Architecture	Encoder backbone	IMPALA	IMPALA	IMPALA
	Optical-flow encoder	RAFT	—	—
	World-model scale	24	—	—
	IDM encoder scale	4	—	—
	Policy encoder scale	—	4	4
	Latent action dimension d_z	128	128	128
	Hidden dim (actor)	—	—	256
LAOF-specific	Future obs. offset	1	—	—

Table 8: Per-stage hyperparameters for LAPO-Slots [14] (see Section B for the pipeline description). Stage 2 (slot probing) has no learnable parameters.

	Hyperparameter	Stage 1: VideoSAUR	Stage 2: Probing	Stage 3: LAM
Optimization	Optimizer	AdamW	—	AdamW
	Learning rate	3×10^{-4}	—	3×10^{-4} / 1.5×10^{-4}
	LR schedule	exp. + warmup	—	—
	Warmup	1,250 steps	—	0 epochs
	Batch size	256 / 128	—	256 / 128
	Gradient clipping	0.05	—	1.0
Training budget	Epochs / steps	25,000 steps	—	1 / 4 epochs
	Dataset subset	full	—	full
Architecture	DINO backbone	ViT-B/14 (DINOv2)	—	ViT-B/14 (DINOv2)
	Slot dim	128	—	128
	Num. slots	4 / 3	—	4 / 3
	Slot iterations	3	—	—
	Sim. loss / temp.	0.1 / 0.075	—	—
	Hidden dim	—	—	1024
	Residual blocks	—	—	3
	Latent action dimension d_z	—	—	128
	Frame stack	—	—	1
LAPO-Slots-specific	PCA components	—	32	—
	Probing model	—	Linear regression	—
	Cross-validation	—	5-fold	—
	Future obs. offset	—	—	10

	Hyperparameter	Stage 4: BC	Stage 5: Decoder
Optimization	Optimizer	AdamW	AdamW
	Learning rate	3×10^{-4} / 1.5×10^{-4}	1×10^{-3}
	LR schedule	—	—
	Warmup	0 epochs	0 epochs
	Batch size	256 / 128	256 / 128
	Gradient clipping	1.0	—
Training budget	Epochs / steps	1 / 4 epochs	25,000 steps
	Dataset subset	128,000	128,000
Architecture	Visual encoder	IMPALA	IMPALA
	Channel multiplier	4	—
	Residual blocks	2	—
	Latent action dimension d_z	128	128
	Hidden dim (actor)	256	256

D Expert Results

We report the performance of the expert policies used to collect the offline trajectories that train all latent action models in this work (Section Q). Table 9 lists the average undiscounted episode return of each DCS expert, and Table 10 lists the average task success rate of each DMW expert, both measured over the same evaluation protocol used for the downstream policies (Section E). These numbers serve as the upper reference point when computing the Normalized Return (NR) on DCS and the Normalized Success Rate (NSR) on DMW: a value of 1.0 matches expert performance on a given task, and 0.0 corresponds to a random policy. Reporting expert results explicitly makes the absolute scale of our normalized metrics interpretable and ensures that any apparent gap to expert performance reflects the difficulty of the task rather than an unattainable normalization target.

Table 9: Expert returns of data collection policy used for normalization on the DCS.

Task	Return
cheetah-run	837.67
hopper-hop	307.33
humanoid-walk	616.52
walker-run	738.37

Table 10: Expert success rates of data collection policy used for normalization on DMW.

Task	Success Rate
button-press-topdown	1.000
door-open	0.917
drawer-close	1.000
drawer-open	1.000
pick-place	1.000
peg-insert-side	0.772
push	1.000
reach	1.000
window-open	1.000
window-close	1.000

E Evaluation Details

All numbers reported in the paper are computed on the held-out evaluation split of the released datasets (Section Q), averaged over three random seeds, with 100 rollout episodes per seed for downstream performance. The remainder of this section makes that protocol explicit, gives the formal definitions of the three metrics summarized in Section 5, and lists the per-task action variances that normalize the probe MSE; hardware and wall-clock times are reported in Section B.

Evaluation data. Every metric is computed on the held-out split of the released dataset (Section Q): 1M transitions on DCS and 100k on DMW. Held-out trajectories sample distractor backgrounds exclusively from the DAVIS test split, while training trajectories sample exclusively from the DAVIS train split, so the two distractor-video sets are disjoint at the clip level. LAOM [20] reports two numbers per task, an in-distribution number that reuses training DAVIS clips at evaluation time and a separate, harder "OOD" number on held-out clips; every number we report corresponds to that OOD setting. We treat this as the only fair protocol, since an in-distribution evaluation conflates memorization of the background videos with generalization to distractor variation, which is the property the method is supposed to provide.

Action clipping. As stated in Section 5, we clip ground-truth actions to the environment action space $[-1, 1]^{d_a}$ prior to probe regression and behavior-cloning fine-tuning. The simulator already clamps commands at execution time, so any two distinct unclipped values that share a clipped image produce identical state transitions; regressing against unclipped targets therefore introduces a many-to-one ambiguity that penalizes models that correctly recover the executed action. Our action-probe MSE values are consequently not directly comparable to LAOM [20], which evaluates on unclipped targets.

Normalized linear action probe MSE (NMSE). Let d_a denote the ground-truth action dimension and d_z the latent action dimension. For each task τ we fit a linear probe consisting of a weight matrix $W \in \mathbb{R}^{d_a \times d_z}$ and a bias vector $b \in \mathbb{R}^{d_a}$ that maps a frozen latent action $z_t \in \mathbb{R}^{d_z}$ to the clipped ground-truth action $a_t \in \mathbb{R}^{d_a}$, with no gradient flowing back into the latent action model. The probe is fit on the training split and evaluated on the held-out split; we report

$$\mathrm{NMSE}(\tau) \;=\; \frac{\mathbb{E}_{\mathcal{D}^{\tau}_{\mathrm{eval}}}\left[\|W z_t + b - a_t\|_2^2\right]}{\mathrm{Var}_{\mathcal{D}^{\tau}}(a)}, \qquad \mathrm{Var}(a) \;=\; \frac{1}{d_a}\sum_{i=1}^{d_a} \mathrm{Var}\left(a^{(i)}\right), \tag{5}$$

where the denominator is the mean per-dimension variance of the clipped ground-truth actions on that task's training split, listed in Table 11. Under this normalization, $\mathrm{NMSE} = 1$ corresponds to a probe that predicts the mean action and $\mathrm{NMSE} = 0$ to perfect recovery, so the metric is comparable across tasks with different action scales.

Normalized return (NR). On DCS we generate rollouts with the decoded policy of Stage 3 for 100 episodes and report

$$\mathrm{NR}(\tau) \;=\; \frac{R_{\pi}(\tau)}{R_{\mathrm{exp}}(\tau)}, \tag{6}$$

where $R_{\pi}(\tau)$ is the mean undiscounted episode return of the learned policy and $R_{\mathrm{exp}}(\tau)$ is the expert return from Table 9. A value of 1 matches the data-collection expert.

Normalized success rate (NSR). On DMW we generate rollouts with the decoded policy of Stage 3 for 100 episodes and report

$$\mathrm{NSR}(\tau) \;=\; \frac{S_{\pi}(\tau)}{S_{\mathrm{exp}}(\tau)}, \tag{7}$$

where $S_{\pi}(\tau)$ is the success fraction of the learned policy and $S_{\mathrm{exp}}(\tau)$ the expert success rate from Table 10.

Aggregation. For every task we first compute the mean and standard deviation across the three seeds. Per-benchmark numbers are obtained by averaging the per-task means, and separately averaging the per-task standard deviations, across the tasks of that benchmark. When we report a single number across both benchmarks, we average the two per-benchmark aggregates, so that DCS and DMW contribute equally regardless of the difference in the number of tasks.

Per-task action variance. Table 11 lists $\mathrm{Var}(a)$ for every evaluated task, which serves as the denominator of (5). Variances are computed on the clipped ground-truth actions of the released dataset and agree to numerical precision between the training and held-out splits, so the split subscript is dropped in (5); we report the value computed on the larger training split. They are also shared between the distractor-free and distracting variants of each benchmark, since the action stream is identical by construction in our synchronized data collection (Section Q).

Table 11: Per-task action variance $\mathrm{Var}(a) = \frac{1}{d_a}\sum_i \mathrm{Var}(a^{(i)})$, computed on clipped ground-truth actions over the training split of the released dataset. Values are shared between the distractor-free and distracting variants of each benchmark, because the action stream is identical by construction (Section Q). Used as the denominator of NMSE in (5).

Task	$\mathrm{Var}(a)$
DCS	
cheetah-run	0.6787
hopper-hop	0.7454
humanoid-walk	0.4221
walker-run	0.6729
DMW	
button-press-topdown	0.2085
door-open	0.4266
drawer-close	0.2466
drawer-open	0.1483
peg-insert-side	0.2497
pick-place	0.2286
push	0.1311
reach	0.0372
window-close	0.3650
window-open	0.2979

F Failure Modes of LAOF and LAPO-Slots

Figure 10: LAOF flow target construction on DCS cheetah-run (distractor). Eight transitions sampled from the dataset. Rows top-to-bottom: current observation o_t; next observation o_{t+1}; supervisory mask M_{t+1}; raw RAFT optical flow between o_t and o_{t+1}; masked flow $M_{t+1} \odot \mathrm{RAFT}(o_t, o_{t+1})$, the target for LAOF's auxiliary flow decoder loss. Distractor motion in the dynamic background produces large flow magnitudes that bleed into the cheetah mask along its boundary, so the masked target inside the mask is dominated by exogenous motion rather than agent motion. The flow decoder is consequently asked to predict distractor flow conditioned on z_t, which pushes z_t to encode it.

On our distractor benchmarks, LAOF and LAPO-Slots perform on par with or only marginally improve over LAPO (Table 1), in contrast to the strong results reported in the original papers. To

explain this gap, we trace each method back to its mask-based prior and identify a distinct failure mode in each case. Both could in principle be alleviated by replacing the underlying component, the optical flow model for LAOF and the object-centric encoder for LAPO-Slots, with a stronger alternative; the conclusions here concern the priors as currently instantiated, not the methods in their idealized form.

LAOF: distractor optical flow bleeds into the agent mask. LAOF supervises an auxiliary flow decoder on RAFT optical flow restricted to the agent mask, so that only agent-driven motion shapes the latent action z_t. Figure 10 shows the construction on DCS cheetah-run with distractors. Distractor regions adjacent to the cheetah produce large RAFT flow magnitudes that, after multiplying by M_{t+1}, contaminate the target inside the mask, so the bottom-row flow the decoder is asked to predict is mostly exogenous motion. The decoder must therefore explain that motion conditioned on z_t, which forces z_t to carry distractor information just as in unmasked LAPO. Figure 11 shows the same construction on DMW push with distractors. The bleed covers a smaller area, but not because the mask is sharper: at DMW's higher rendering resolution the agent simply occupies more pixels, so the contaminated boundary region is a smaller fraction of the masked area. The distractor flow magnitude remains high relative to the slow robot arm, so wherever the bleed does occur it still dominates the local target.

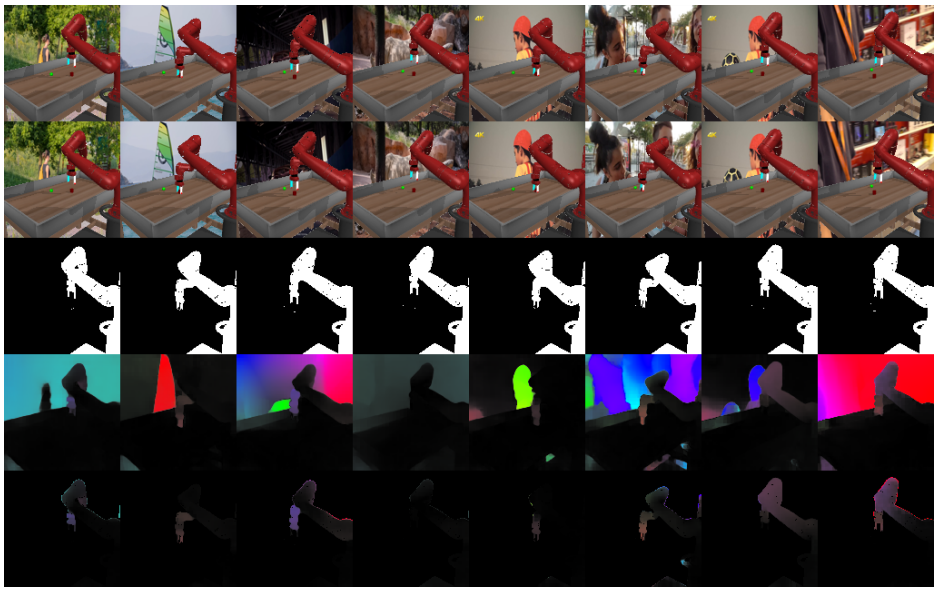

Figure 11: LAOF flow target construction on DMW push (distractor). Layout matches Figure 10. The contaminated boundary region covers a smaller fraction of the masked area than on DCS, not because the mask is sharper but because the agent occupies more pixels at DMW's higher rendering resolution. The distractor flow magnitude is still high relative to the slow robot arm, so where bleed does occur it still dominates the local target.

LAPO-Slots: agent identity is not stable across slot indices. The full LAPO-Slots pipeline is described in Section B; here we focus on the three stages relevant to the failure mode. Stage 1 (VideoSAUR) decomposes each frame into K object-centric slots; Stage 2 selects the single most action-predictive slot index per task by linearly probing each slot's features against ground-truth actions; Stage 3 trains the LAM on the features of that one selected slot. The pipeline therefore depends on the agent occupying the same slot index across all samples in the dataset, since the Stage 2 selection is a single fixed index applied uniformly at Stage 3 training time: whenever VideoSAUR routes the agent to a different slot on a sample, the LAM input on that sample contains no agent. We use $K{=}4$ slots on DCS and $K{=}3$ slots on DMW, following the original protocol [14].

Figure 12 shows a representative *stable* run on DCS cheetah-run (distractor): slot 2 attends to the cheetah across all eight sampled transitions while the remaining slots absorb background structure, so a Stage 2 selection of slot 2 retains the agent on every sample. Figure 13 shows the failure mode on the same task: cheetah identity is not bound to a single slot index, but is split across slot 3 and

slot 4 across the sampled batch, so no fixed slot index covers the agent on every sample. Figures 14 and 15 show the same pattern on DMW push (distractor): a stable run where slot 2 carries the robot arm on every sample, and a failed run where arm identity is distributed across slots 1, 2, and 3 across the sampled batch. Because Stage 3 commits to a single slot index, this instability translates directly into LAM input dropout on the samples where VideoSAUR routes the agent elsewhere.

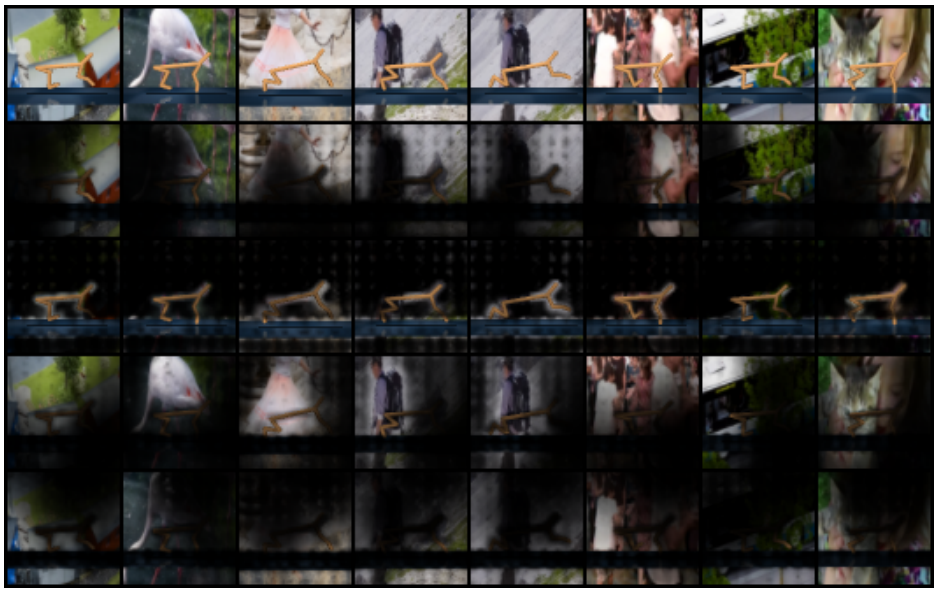

Figure 12: LAPO-Slots VideoSAUR slot decomposition on DCS cheetah-run (distractor), stable run. Eight transitions sampled from the dataset. Top row: input observation o_t. Rows 2–5: per-slot masked observations for slots 1–4. Slot 2 (row 3) consistently attends to the cheetah on every sampled frame; the remaining slots absorb distractor background structure. A Stage 2 selection of slot 2 therefore retains the agent on every dataset sample and the downstream LAM input is well defined.

Figure 13: LAPO-Slots VideoSAUR slot decomposition on DCS cheetah-run (distractor), failed run. Layout matches Figure 12. Cheetah identity is split between slots 3 and 4 (rows 4 and 5) across the sampled batch, so no single slot index covers the agent on every sample. Because Stage 3 trains the LAM on the features of one fixed slot, the agent is dropped from the LAM input on the samples where VideoSAUR assigns it to the other slot.

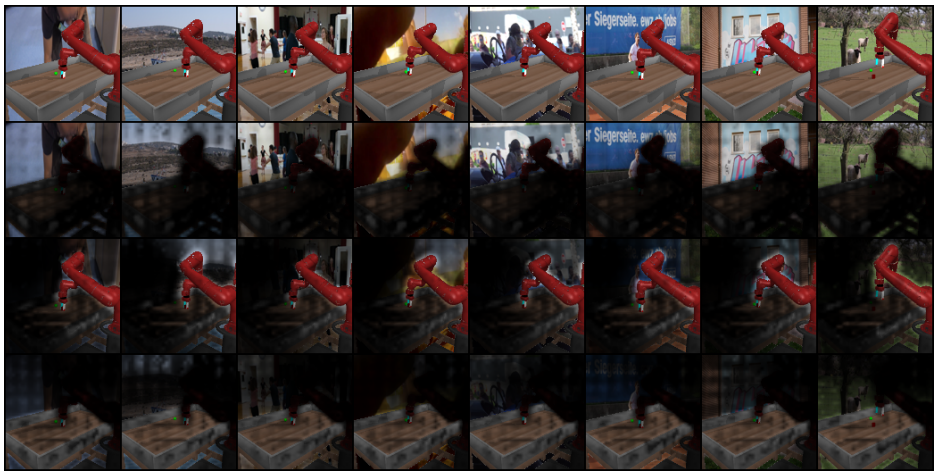

Figure 14: LAPO-Slots VideoSAUR slot decomposition on DMW push (distractor), stable run. Eight transitions sampled from the dataset. Top row: input observation o_t. Rows 2–4: per-slot masked observations for slots 1–3. Slot 2 (row 3) consistently attends to the robot arm on every sampled frame; slots 1 and 3 absorb table and distractor regions.

Figure 15: LAPO-Slots VideoSAUR slot decomposition on DMW push (distractor), failed run. Layout matches Figure 14. Robot-arm identity is distributed across slots 1, 2, and 3 across the sampled batch, so any fixed Stage 2 slot selection loses the arm on a subset of samples and the downstream LAM is trained on intermittently empty input.

G World Model Visualizations

The FDM (world model) in Stage 1 of the three-stage MaskLAM and LAPO training pipeline is trained jointly with the IDM (Section 4) and provides the only learning signal that shapes the latent action z_t: z_t must explain whatever the world model is asked to reconstruct beyond what the prior frames already determine. Visualizing what each method's world model actually predicts therefore exposes what z_t has been forced to encode.

Figures 16 and 17 show LAPO's world model on DCS (distractor) and DMW (distractor). Reconstructions are nearly indistinguishable from the ground-truth next observation: the agent, the dynamic distractor video behind it, the camera shake, and the color randomization are all recovered at the pixel level. This is exactly what the unmasked reconstruction loss rewards. The world model has no way to know that the distractor video is exogenous, so it spends representational capacity on tracking it, and the IDM is pushed to write frame-to-frame distractor variation that the prior frames cannot predict

into z_t. The pixel-perfect reconstruction is direct visual evidence of the failure mode that motivates our method.

Figures 18 and 19 show MaskLAM's world model on the same benchmarks. The agent and its immediate vicinity are reconstructed cleanly, while everything outside the supervisory mask collapses into a low-frequency texture with visible artifacts at the agent boundary. This is the intended behavior: with the loss masked to M_{t+1}, the world model receives no gradient for distractor pixels and stops modeling them, so z_t is no longer required to carry distractor information. The artifacts outside the mask are not a defect but the visible signature of the bottleneck we impose on the latent action.

Figure 16: LAPO world model predictions on DCS (distractor). Top row: predicted next observation $\hat{o}_{t+1}$. Bottom row: ground-truth next observation o_{t+1}. Ten consecutive transitions sampled from training. The unmasked reconstruction loss yields pixel-accurate reproduction of the dynamic distractor video, camera shake, and agent color randomization, evidence that z_t must carry distractor information for the world model to satisfy its objective.

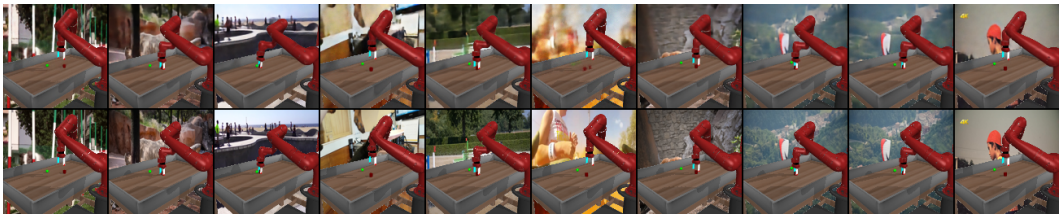

Figure 17: LAPO world model predictions on DMW (distractor). Layout matches Figure 16. Reconstructions track the distractor background with the same fidelity as the agent, confirming that the failure mode is benchmark-independent.

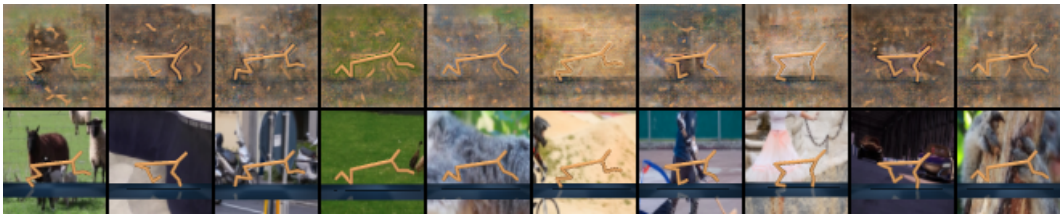

Figure 18: MaskLAM world model predictions on DCS (distractor). Top row: predicted next observation $\hat{o}_{t+1}$. Bottom row: ground-truth next observation o_{t+1}. With the reconstruction loss restricted to the supervisory mask M_{t+1}, the world model recovers the agent sharply while everything outside the mask collapses into a low-frequency texture with boundary artifacts.

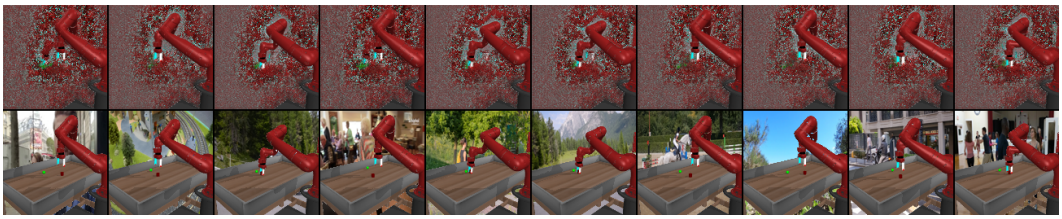

Figure 19: MaskLAM world model predictions on DMW (distractor). Layout matches Figure 18. The agent and its contact surfaces are reconstructed cleanly; the masked-out background is reconstructed only to a coarse extent, mirroring the DCS behavior.

H Masking Improves Latent Action Quality

Per-task breakdown of the aggregated NMSE in Table 1; metric definition in Section E.

Distractor setting. MaskLAM@GT and MaskLAM@SAM attain the lowest NMSE among label-free baselines on all four DCS tasks (Table 12) and all ten DMW tasks (Table 14), often by a 2–3× margin, and match LAOM-Labels on the majority of tasks. The two MaskLAM variants agree within one standard deviation on every task.

Distractor-free setting. On DCS (Table 12, columns 5–8), MaskLAM matches or exceeds every label-free baseline and lands near the privileged LAOM-Labels. On DMW (Table 13), MaskLAM ties LAPO, which is the strongest baseline in the absence of distractors. Without exogenous variance dominating the reconstruction target, the unmasked loss already places most gradient weight on agent-driven change.

Distracting Control Suite (NMSE ↓)

	Distractor				Vanilla			
Method	**cheetah-run**	**hopper-hop**	**humanoid-walk**	**walker-run**	**cheetah-run**	**hopper-hop**	**humanoid-walk**	**walker-run**
LAPO	$0.3836_{\pm 0.0220}$	$0.6433_{\pm 0.0671}$	$0.6766_{\pm 0.0382}$	$0.3423_{\pm 0.0071}$	$0.1484_{\pm 0.0019}$	$0.1959_{\pm 0.0036}$	$0.5298_{\pm 0.2591}$	$0.3142_{\pm 0.2031}$
LAOM-Labels	$0.1652_{\pm 0.0020}$	$0.1933_{\pm 0.0067}$	$\mathbf{0.3112_{\pm 0.0025}}$	$0.1943_{\pm 0.0026}$	$0.1478_{\pm 0.0009}$	$\mathbf{0.1904_{\pm 0.0156}}$	$\mathbf{0.3379_{\pm 0.0072}}$	$\mathbf{0.1663_{\pm 0.0006}}$
LAOF	$0.3236_{\pm 0.0143}$	$0.4863_{\pm 0.0303}$	$0.5719_{\pm 0.0174}$	$0.6408_{\pm 0.0157}$	$0.5546_{\pm 0.0162}$	$0.7345_{\pm 0.0443}$	$0.6322_{\pm 0.0503}$	$0.6348_{\pm 0.0696}$
LAPO-Slots	$0.6615_{\pm 0.0212}$	$0.6950_{\pm 0.0051}$	$0.7270_{\pm 0.0052}$	$0.5444_{\pm 0.0859}$	$0.2892_{\pm 0.0314}$	$0.3613_{\pm 0.0066}$	$0.4673_{\pm 0.0088}$	$0.2884_{\pm 0.0020}$
MaskLAM@GT	$0.1171_{\pm 0.0007}$	$\mathbf{0.1853_{\pm 0.0256}}$	$0.3531_{\pm 0.0059}$	$\mathbf{0.1692_{\pm 0.0049}}$	$0.0910_{\pm 0.0017}$	$0.2093_{\pm 0.0031}$	$0.3380_{\pm 0.0041}$	$0.1811_{\pm 0.0163}$
MaskLAM@SAM	$\mathbf{0.1148_{\pm 0.0040}}$	$0.2601_{\pm 0.1434}$	$0.3852_{\pm 0.0007}$	$0.1704_{\pm 0.0008}$	$\mathbf{0.0859_{\pm 0.0028}}$	$0.2147_{\pm 0.0245}$	$0.3555_{\pm 0.0076}$	$0.1956_{\pm 0.0035}$

Table 12: Per-task NMSE on DCS in distractor (columns 1–4) and distractor-free (columns 5–8) settings. MaskLAM@GT and MaskLAM@SAM attain the lowest NMSE among label-free baselines under distractors and surpass LAOM-Labels on three of four tasks; without distractors MaskLAM matches or exceeds every label-free baseline and lands within noise of LAOM-Labels.

Distracting Meta-World - Vanilla (NMSE ↓)

Method	**reach**	**push**	**pick-place**	**door-open**	**drawer-open**	**drawer-close**	**button-press-topdown**	**peg-insert-side**	**window-open**	**window-close**
LAPO	$\mathbf{0.0773_{\pm 0.0086}}$	$0.1332_{\pm 0.0076}$	$\mathbf{0.1105_{\pm 0.0015}}$	$\mathbf{0.0557_{\pm 0.0009}}$	$\mathbf{0.0399_{\pm 0.0018}}$	$0.0124_{\pm 0.0012}$	$0.0358_{\pm 0.0034}$	$0.2174_{\pm 0.0068}$	$0.0466_{\pm 0.0024}$	$\mathbf{0.0129_{\pm 0.0014}}$
LAOM-Labels	$0.1130_{\pm 0.0120}$	$0.1583_{\pm 0.0146}$	$0.1273_{\pm 0.0073}$	$0.1047_{\pm 0.0103}$	$0.0986_{\pm 0.0193}$	$0.0931_{\pm 0.0065}$	$0.1068_{\pm 0.0097}$	$0.1983_{\pm 0.0100}$	$0.0789_{\pm 0.0034}$	$0.0584_{\pm 0.0058}$
LAOF	$0.1254_{\pm 0.0049}$	$0.3574_{\pm 0.0437}$	$0.2276_{\pm 0.0235}$	$0.2055_{\pm 0.1099}$	$0.1483_{\pm 0.0398}$	$0.1249_{\pm 0.0104}$	$0.1680_{\pm 0.0855}$	$0.3118_{\pm 0.0223}$	$0.1192_{\pm 0.0191}$	$0.0605_{\pm 0.0113}$
LAPO-Slots	$0.1117_{\pm 0.0046}$	$0.2166_{\pm 0.0118}$	$0.1675_{\pm 0.0049}$	$0.1245_{\pm 0.0081}$	$0.0757_{\pm 0.0031}$	$0.0540_{\pm 0.0041}$	$0.0447_{\pm 0.0031}$	$\mathbf{0.1979_{\pm 0.0095}}$	$0.0678_{\pm 0.0053}$	$0.0292_{\pm 0.0058}$
MaskLAM@GT	$0.0837_{\pm 0.0063}$	$0.1376_{\pm 0.0047}$	$0.1290_{\pm 0.0093}$	$0.0763_{\pm 0.0017}$	$0.0439_{\pm 0.0019}$	$0.0137_{\pm 0.0026}$	$0.0341_{\pm 0.0018}$	$0.2417_{\pm 0.0079}$	$\mathbf{0.0399_{\pm 0.0014}}$	$0.0155_{\pm 0.0030}$
MaskLAM@SAM	$0.0794_{\pm 0.0057}$	$\mathbf{0.1271_{\pm 0.0018}}$	$0.1259_{\pm 0.0041}$	$0.0667_{\pm 0.0029}$	$0.0468_{\pm 0.0021}$	$\mathbf{0.0108_{\pm 0.0008}}$	$\mathbf{0.0292_{\pm 0.0021}}$	$0.2354_{\pm 0.0056}$	$0.0448_{\pm 0.0038}$	$0.0131_{\pm 0.0012}$

Table 13: Per-task NMSE on DMW (vanilla). MaskLAM ties LAPO, the strongest label-free baseline without distractors, confirming that loss masking does not degrade alignment in the absence of exogenous variance.

Distracting Meta-World - Distractor (NMSE ↓)

Method	reach	push	pick-place	door-open	drawer-open	drawer-close	button-press-topdown	peg-insert-side	window-open	window-close
LAPO	$0.2906_{\pm 0.0087}$	$0.4812_{\pm 0.0028}$	$0.5470_{\pm 0.0110}$	$0.2312_{\pm 0.0063}$	$0.2612_{\pm 0.0087}$	$0.1715_{\pm 0.0067}$	$0.2978_{\pm 0.0131}$	$0.5117_{\pm 0.0176}$	$0.2021_{\pm 0.0140}$	$0.0987_{\pm 0.0051}$
LAOM-Labels	$0.1610_{\pm 0.0027}$	$0.2065_{\pm 0.0060}$	$0.1409_{\pm 0.0030}$	$0.1407_{\pm 0.0124}$	$0.1143_{\pm 0.0183}$	$0.1335_{\pm 0.0138}$	$0.1075_{\pm 0.0029}$	$0.2496_{\pm 0.0065}$	$0.0916_{\pm 0.0067}$	$0.0751_{\pm 0.0050}$
LAOF	$0.3392_{\pm 0.0244}$	$0.5003_{\pm 0.0645}$	$0.4891_{\pm 0.0497}$	$0.4487_{\pm 0.0802}$	$0.4893_{\pm 0.1146}$	$0.5076_{\pm 0.0681}$	$0.4376_{\pm 0.0176}$	$0.5331_{\pm 0.0436}$	$0.4662_{\pm 0.0529}$	$0.3180_{\pm 0.0530}$
LAPO-Slots	$0.2174_{\pm 0.0049}$	$0.4550_{\pm 0.2032}$	$0.2691_{\pm 0.0081}$	$0.2444_{\pm 0.0319}$	$0.2927_{\pm 0.0616}$	$0.1666_{\pm 0.0047}$	$0.2083_{\pm 0.1377}$	$0.3184_{\pm 0.0070}$	$0.1600_{\pm 0.0045}$	$0.1183_{\pm 0.0179}$
MaskLAM@GT	$\mathbf{0.1140_{\pm 0.0066}}$	$0.1441_{\pm 0.0080}$	$\mathbf{0.1291_{\pm 0.0005}}$	$\mathbf{0.0738_{\pm 0.0077}}$	$0.0548_{\pm 0.0021}$	$\mathbf{0.0156_{\pm 0.0003}}$	$\mathbf{0.0391_{\pm 0.0018}}$	$0.2538_{\pm 0.0068}$	$\mathbf{0.0409_{\pm 0.0008}}$	$\mathbf{0.0148_{\pm 0.0016}}$
MaskLAM@SAM	$0.1246_{\pm 0.0077}$	$\mathbf{0.1366_{\pm 0.0061}}$	$0.1316_{\pm 0.0077}$	$0.0789_{\pm 0.0068}$	$\mathbf{0.0524_{\pm 0.0046}}$	$0.0184_{\pm 0.0015}$	$0.0462_{\pm 0.0007}$	$\mathbf{0.2381_{\pm 0.0100}}$	$0.0605_{\pm 0.0022}$	$0.0161_{\pm 0.0018}$

Table 14: Per-task NMSE on DMW (distractor). MaskLAM@GT and MaskLAM@SAM reduce NMSE by 2–3× relative to LAPO and surpass every other baseline on all ten tasks, including the contact-rich pick-place, peg-insert-side, and push.

I Masking Improves Downstream Returns

Per-task breakdown of the aggregated downstream return in Table 1; metric definitions in Section E. All numbers are mean ± standard deviation across three seeds.

Distractor setting. MaskLAM@GT and MaskLAM@SAM exceed every label-free baseline on every DCS task (Table 15), and match or exceed LAOM-Labels on 7 of 10 DMW tasks (Table 17) despite using no action supervision in Stage 1.

Distractor-free setting. On DCS (Table 15, columns 5–8), LAOM-Labels is the top-performing method, but MaskLAM stays in reach. On DMW (Table 16), MaskLAM ties LAPO, which is the strongest baseline in the absence of distractors.

Distracting Control Suite (Mean NR ↑)

	Distractor				Vanilla			
Method	cheetah-run	hopper-hop	humanoid-walk	walker-run	cheetah-run	hopper-hop	humanoid-walk	walker-run
LAPO	$0.2057_{\pm 0.0211}$	$0.0043_{\pm 0.0005}$	$0.0375_{\pm 0.0115}$	$0.0562_{\pm 0.0038}$	$0.6828_{\pm 0.0431}$	$0.1729_{\pm 0.0180}$	$0.0022_{\pm 0.0001}$	$0.0876_{\pm 0.0496}$
LAOM-Labels	$\mathbf{0.9415_{\pm 0.0018}}$	$\mathbf{0.5425_{\pm 0.0120}}$	$0.0448_{\pm 0.0030}$	$\mathbf{0.7364_{\pm 0.0578}}$	$0.9219_{\pm 0.0091}$	$\mathbf{0.7114_{\pm 0.0292}}$	$0.0019_{\pm 0.0001}$	$\mathbf{0.7981_{\pm 0.0275}}$
LAOF	$0.2721_{\pm 0.0529}$	$0.0561_{\pm 0.0189}$	$0.0082_{\pm 0.0022}$	$0.0391_{\pm 0.0040}$	$0.2131_{\pm 0.0403}$	$0.0281_{\pm 0.0155}$	$0.0022_{\pm 0.0000}$	$0.0452_{\pm 0.0119}$
LAPO-Slots	$0.1055_{\pm 0.0141}$	$0.0020_{\pm 0.0007}$	$0.0247_{\pm 0.0052}$	$0.0494_{\pm 0.0113}$	$0.1801_{\pm 0.0146}$	$0.1211_{\pm 0.0234}$	$0.0020_{\pm 0.0002}$	$0.2023_{\pm 0.0437}$
MaskLAM@GT	$0.7588_{\pm 0.0309}$	$0.2736_{\pm 0.0382}$	$\mathbf{0.0758_{\pm 0.0121}}$	$0.4022_{\pm 0.0381}$	$\mathbf{0.9362_{\pm 0.0286}}$	$0.6337_{\pm 0.0451}$	$0.0024_{\pm 0.0000}$	$0.4782_{\pm 0.0407}$
MaskLAM@SAM	$0.7384_{\pm 0.0342}$	$0.2164_{\pm 0.1050}$	$0.0645_{\pm 0.0086}$	$0.4130_{\pm 0.0136}$	$0.9167_{\pm 0.0151}$	$0.5628_{\pm 0.0733}$	$\mathbf{0.0024_{\pm 0.0002}}$	$0.4919_{\pm 0.0548}$

Table 15: Per-task NR on DCS in distractor (columns 1–4) and distractor-free (columns 5–8) settings. MaskLAM is the top-performing label-free method on every task under distractors and on 3 of 4 tasks without distractors, where it also lands near LAOM-Labels.

Distracting Meta-World - Vanilla (Mean NSR ↑)

Method	reach	push	pick-place	door-open	drawer-open	drawer-close	button-press-topdown	peg-insert-side	window-open	window-close
LAPO	$0.5648_{\pm 0.1118}$	$\mathbf{0.9364_{\pm 0.0348}}$	$0.7194_{\pm 0.0555}$	$0.9212_{\pm 0.0212}$	$\mathbf{1.0000_{\pm 0.0000}}$	$\mathbf{1.0000_{\pm 0.0000}}$	$\mathbf{1.0000_{\pm 0.0000}}$	$0.4011_{\pm 0.0187}$	$0.9668_{\pm 0.0146}$	$\mathbf{1.0000_{\pm 0.0000}}$
LAOM-Labels	$0.6133_{\pm 0.0633}$	$0.1983_{\pm 0.0451}$	$0.0750_{\pm 0.0328}$	$0.9400_{\pm 0.0350}$	$\mathbf{1.0000_{\pm 0.0000}}$	$0.9967_{\pm 0.0058}$	$0.9900_{\pm 0.0000}$	$\mathbf{0.4567_{\pm 0.0961}}$	$\mathbf{1.0000_{\pm 0.0000}}$	$\mathbf{1.0000_{\pm 0.0000}}$
LAOF	$\mathbf{0.7297_{\pm 0.2191}}$	$0.4307_{\pm 0.2137}$	$0.3347_{\pm 0.1592}$	$\mathbf{0.9690_{\pm 0.0494}}$	$\mathbf{1.0000_{\pm 0.0000}}$	$\mathbf{1.0000_{\pm 0.0000}}$	$0.8317_{\pm 0.2916}$	$0.4140_{\pm 0.0233}$	$0.4507_{\pm 0.2192}$	$\mathbf{1.0000_{\pm 0.0000}}$
LAPO-Slots	$0.5076_{\pm 0.0798}$	$0.9242_{\pm 0.0284}$	$0.6704_{\pm 0.0571}$	$0.9468_{\pm 0.0150}$	$\mathbf{1.0000_{\pm 0.0000}}$	$\mathbf{1.0000_{\pm 0.0000}}$	$\mathbf{1.0000_{\pm 0.0000}}$	$0.2811_{\pm 0.1385}$	$0.9963_{\pm 0.0042}$	$\mathbf{1.0000_{\pm 0.0000}}$
MaskLAM@GT	$0.5430_{\pm 0.0188}$	$0.9171_{\pm 0.0379}$	$0.7054_{\pm 0.0556}$	$0.9213_{\pm 0.0389}$	$\mathbf{1.0000_{\pm 0.0000}}$	$\mathbf{1.0000_{\pm 0.0000}}$	$\mathbf{1.0000_{\pm 0.0000}}$	$0.3102_{\pm 0.0812}$	$0.9507_{\pm 0.0304}$	$0.9281_{\pm 0.1012}$
MaskLAM@SAM	$0.5375_{\pm 0.0591}$	$0.9143_{\pm 0.0100}$	$\mathbf{0.7220_{\pm 0.0368}}$	$0.9378_{\pm 0.0161}$	$\mathbf{1.0000_{\pm 0.0000}}$	$\mathbf{1.0000_{\pm 0.0000}}$	$\mathbf{1.0000_{\pm 0.0000}}$	$0.2369_{\pm 0.0570}$	$0.9372_{\pm 0.0324}$	$0.9958_{\pm 0.0039}$

Table 16: Per-task NSR on DMW (vanilla). MaskLAM ties LAPO, the strongest baseline in the absence of distractors, indicating that loss masking carries no penalty when its target assumption (exogenous contamination) is absent.

Distracting Meta-World - Distractor (Mean NSR ↑)

Method	reach	push	pick-place	door-open	drawer-open	drawer-close	button-press-topdown	peg-insert-side	window-open	window-close
LAPO	$0.2100_{\pm 0.0211}$	$0.1715_{\pm 0.0461}$	$0.0560_{\pm 0.0207}$	$0.9118_{\pm 0.0247}$	$0.9979_{\pm 0.0029}$	$\mathbf{1.0000_{\pm 0.0000}}$	$0.9934_{\pm 0.0080}$	$0.0535_{\pm 0.0140}$	$0.9698_{\pm 0.0093}$	$0.9981_{\pm 0.0005}$
LAOM-Labels	$0.5250_{\pm 0.0527}$	$0.5250_{\pm 0.0482}$	$0.4433_{\pm 0.1079}$	$\mathbf{0.9483_{\pm 0.0153}}$	$\mathbf{1.0000_{\pm 0.0000}}$	$\mathbf{1.0000_{\pm 0.0000}}$	$\mathbf{1.0000_{\pm 0.0000}}$	$\mathbf{0.3683_{\pm 0.1040}}$	$\mathbf{1.0000_{\pm 0.0000}}$	$\mathbf{1.0000_{\pm 0.0000}}$
LAOF	$0.3927_{\pm 0.0264}$	$0.3560_{\pm 0.1863}$	$0.2853_{\pm 0.1513}$	$0.9340_{\pm 0.0245}$	$\mathbf{1.0000_{\pm 0.0000}}$	$0.9200_{\pm 0.1213}$	$0.5333_{\pm 0.2197}$	$0.0847_{\pm 0.0133}$	$0.2617_{\pm 0.0480}$	$\mathbf{1.0000_{\pm 0.0000}}$
LAPO-Slots	$0.5321_{\pm 0.0531}$	$0.7736_{\pm 0.1015}$	$0.7330_{\pm 0.0292}$	$0.9124_{\pm 0.0058}$	$0.9965_{\pm 0.0061}$	$\mathbf{1.0000_{\pm 0.0000}}$	$0.9945_{\pm 0.0095}$	$0.3347_{\pm 0.0212}$	$0.9970_{\pm 0.0029}$	$\mathbf{1.0000_{\pm 0.0000}}$
MaskLAM@GT	$\mathbf{0.5451_{\pm 0.0455}}$	$\mathbf{0.9286_{\pm 0.0222}}$	$0.7771_{\pm 0.0511}$	$0.9076_{\pm 0.0319}$	$\mathbf{1.0000_{\pm 0.0000}}$	$\mathbf{1.0000_{\pm 0.0000}}$	$\mathbf{1.0000_{\pm 0.0000}}$	$0.2812_{\pm 0.0542}$	$0.9921_{\pm 0.0062}$	$\mathbf{1.0000_{\pm 0.0000}}$
MaskLAM@SAM	$0.4861_{\pm 0.0460}$	$0.9073_{\pm 0.0469}$	$\mathbf{0.8574_{\pm 0.0293}}$	$0.9133_{\pm 0.0347}$	$\mathbf{1.0000_{\pm 0.0000}}$	$\mathbf{1.0000_{\pm 0.0000}}$	$\mathbf{1.0000_{\pm 0.0000}}$	$0.2966_{\pm 0.0696}$	$0.8578_{\pm 0.0865}$	$0.9994_{\pm 0.0010}$

Table 17: Per-task NSR on DMW (distractor). MaskLAM@GT and MaskLAM@SAM match or exceed LAOM-Labels on 7 of 10 tasks while outperforming every other baseline, despite using no action supervision in Stage 1.

J Masking Improves Sample Efficiency

Per-environment breakdown of Figure 7: NR on DCS (distractor) and NSR on DMW (distractor) as a function of the Stage 3 label budget, swept from 2k to 128k labels with the Stage 1 latent action model held fixed. All numbers in this section are reported in the distractor setting.

DMW. MaskLAM@GT reaches near peak NSR with 2k labels on six of ten tasks (reach, push, pick-place, drawer-open, drawer-close, window-open), while LAPO and LAPO-Slots need nearly two orders of magnitude more labels to close the gap (Figure 20).

DCS. LAPO and LAPO-Slots fail to recover useful policies at any budget we tested; MaskLAM@GT reaches competitive returns from 16k–32k labels (Figure 21). LAOM-Labels remains the ceiling on hopper-hop and walker-run, where well-aligned latents still require more decoder capacity to translate into stable gaits.

Mask source. MaskLAM@SAM tracks MaskLAM@GT across all environments and label budgets, so swapping ground-truth for SAM-predicted masks costs no sample efficiency.

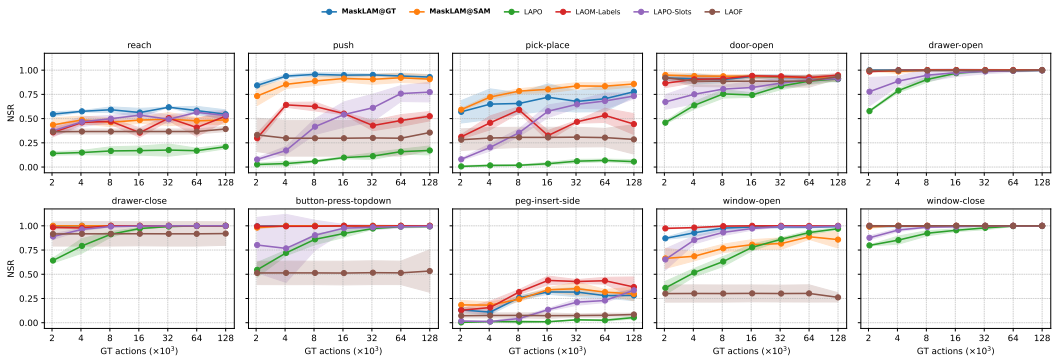

Figure 20: Per-task NSR on DMW (distractor) as a function of the Stage 3 label budget (log scale). MaskLAM reaches peak performance with 2k labels on six of ten tasks; LAPO and LAPO-Slots require nearly two orders of magnitude more labels to close the gap.

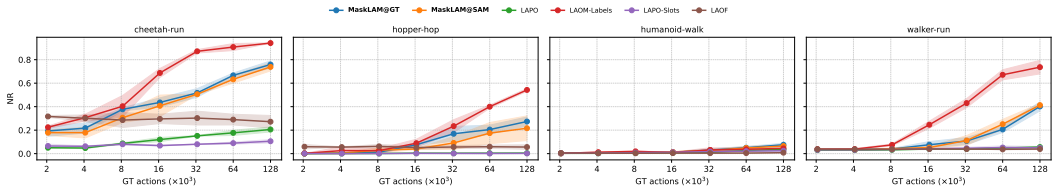

Figure 21: Per-task NR on DCS (distractor) as a function of the Stage 3 label budget (log scale). MaskLAM@GT saturates earlier than every label-free baseline; LAPO and LAPO-Slots fail to recover useful policies at any budget.

K MaskLAM is Robust to Occlusions

Real videos rarely keep the full agent in frame: limbs leave the camera view, foreground objects pass in front of the agent, and contact-rich motion regularly occludes joints behind other parts of the body. A practical latent action model has to keep producing well-aligned latents under this regime. We probe this on the DCS (distractor) by occluding a controlled fraction of the agent at training and evaluation time, and reporting the action-probe NMSE of (5) per task.

Occlusion procedure. For every video sample we synthesize a single random axis-aligned occlusion rectangle anchored to the agent. The construction is as follows:

1. Take the union of the ground-truth foreground mask over the frames of the video sample, $U = \bigvee_t M_t$, and compute its tight axis-aligned bounding box B of area A_B.

2. Sample a target occlusion area $A = \rho\, A_B$, where $\rho \in [0, 1]$ is the occlusion fraction reported on the x-axis of Figure 22.
3. Sample a log-uniform aspect ratio $r \sim \exp(\mathcal{U}(-\frac{1}{2}, \frac{1}{2}))$ and derive height $h = \sqrt{Ar}$ and width $w = A/h$, both clamped to B so the rectangle never leaves the agent footprint.
4. Sample a uniformly random top-left corner inside the remaining slack of B to obtain the final rectangle R.
5. Apply R identically to every frame of the video sample: overwrite the observation pixels inside R with a constant gray fill (corresponding to a 0.0 fill value after normalization) and zero the corresponding entries of the segmentation mask.

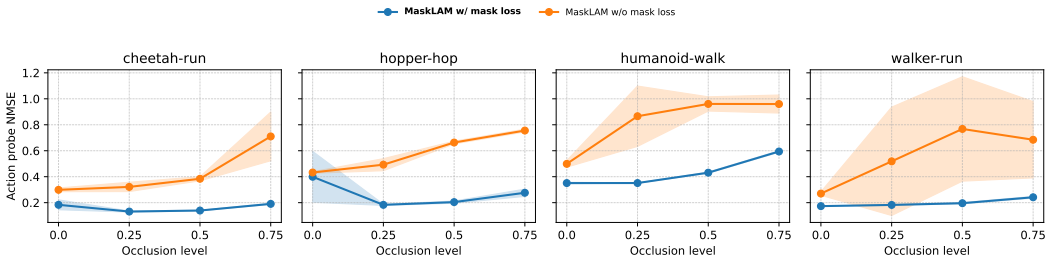

Figure 22: NMSE as a function of agent occlusion fraction on the DCS ($d_z = 128$). The x-axis is the fraction ρ of the agent's bounding-box area occluded by a random gray rectangle at training time (0% = no occlusion, 75% = three-quarters of agent pixels removed). Blue lines: MaskLAM with masked loss; orange lines: same architecture with the masked loss removed. With the masked loss, NMSE stays nearly flat and tightly concentrated across the full occlusion range on all four tasks; without it, error more than doubles and variance increases sharply, because the growing occlusion patch reduces the action-driven fraction of the unmasked reconstruction target while distractor variance remains unchanged.

The rectangle is resampled per trajectory and per training step, so the model never memorizes a fixed occlusion. Zeroing the mask inside R mirrors the behavior of any off-the-shelf segmenter, which simply fails to label pixels it cannot see; the IDM and FDM therefore receive exactly the inputs and supervisory signal they would receive in a deployment where occluded body parts disappear from the predicted segmentation.

Action-probe NMSE under occlusion. Figure 22 sweeps ρ from 0% to 75% for both MaskLAM and an ablation that removes the masked loss. With the masked loss the curve stays nearly flat across the full range on every environment, while removing it more than doubles the probe error on cheetah-run and walker-run and degrades the other two environments by a similar margin. The mask-loss curve is also visibly tighter across seeds.

Why masking helps under occlusion. Occluding the agent does not add new exogenous variance, but it shrinks the visible agent footprint and therefore the action-driven component of the FDM target, while distractor variance behind and around the agent stays the same. Under the unmasked loss the action-to-noise ratio of the supervisory target degrades as ρ grows, and the optimization increasingly pressures z_t to encode the distractor variance that now dominates the residual. This is the linearized PCA picture of Zhang et al. [35] applied to a target whose endogenous content has been thinned: as the agent signal in the target shrinks, the principal direction of frame-to-frame change tilts toward the distractor, and so does z_t. Under the masked loss the occluded rectangle is removed from M_{t+1} before the residual is computed, so the target contains only visible agent pixels at every ρ. The action-to-noise ratio of the supervisory signal stays effectively at one regardless of how much of the agent is hidden, and z_t keeps encoding only action-driven variance. Figures 23 and 24 show this at the pixel level. Without the masked loss the FDM continues to reproduce the full scene, occlusion patch, cheetah, and distractor video, pixel-accurately at every ρ, so z_t keeps carrying all of that variance even as the action-relevant share of the target shrinks. With the masked loss the FDM is asked only to reconstruct visible agent pixels and does so cleanly up to 75% occlusion, while everything outside M_{t+1} collapses to the same low-frequency texture seen in Figure 18. Because zeroing the mask

inside R is exactly what an off-the-shelf segmenter does on partially visible agents, this experiment also demonstrates MaskLAM's robustness to real-world segmentation behavior.

Figure 23: FDM reconstructions of MaskLAM *without* the masked loss objective on cheetah-run (DCS, distractor) under increasing agent occlusion. Rows correspond to occlusion fractions 0%, 25%, 50%, and 75% from top to bottom. Within each row, the top sub-row shows the predicted next observation $\hat{o}_{t+1}$ and the bottom sub-row shows the ground-truth o_{t+1} with the gray occlusion patch overlaid on the cheetah. Without the masked loss, the world model is forced to reconstruct occlusion pixels.

L Masking Enables Compact Latent Action Spaces

Per-environment breakdown of Figure 6a: NMSE on DCS (distractor) as a function of latent action dimension $d_z \in \{32, 64, 128, 256, 512, 1024\}$, with and without the masked loss (all other hyperparameters held fixed).

Compression at fixed alignment. A 256-dim masked latent matches or exceeds a 1024-dim unmasked latent on all four tasks (Figure 25), a $4\times$ reduction with no loss in alignment. The unmasked loss must allocate dimensions to both agent and distractor dynamics; the masked loss restricts the supervisory variance to agent pixels, shrinking the effective rank of the prediction target.

Low-dimensional regime. The gap widens at $d_z \leq 64$, where encoding exogenous content under the unmasked loss starves the action signal entirely.

Figure 24: FDM reconstructions of MaskLAM *with* the masked loss objective on cheetah-run (DCS, distractor) under increasing agent occlusion. Layout matches Figure 23: rows correspond to occlusion fractions 0%, 25%, 50%, and 75%, and within each row the top sub-row is the predicted next observation $\hat{o}_{t+1}$ while the bottom sub-row is the ground-truth o_{t+1} with the gray occlusion patch on the cheetah. Restricting reconstruction to the supervisory mask keeps the prediction inside the mask robust to occlusion, recovering the visible cheetah pixels cleanly up to 75% occlusion and mirroring the action-probe NMSE trend in Figure 22.

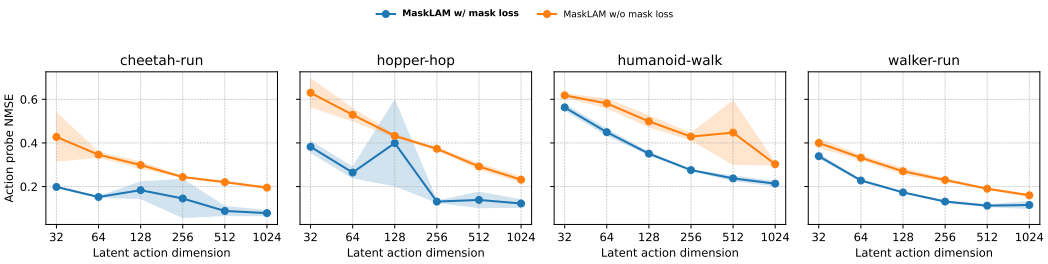

Figure 25: Per-task NMSE on DCS (distractor) as a function of latent action dimension (log scale). Blue: MaskLAM@GT; orange: same architecture with the masked loss removed. The masked loss yields lower NMSE at every d_z. A 256-dim masked model matches a 1024-dim unmasked one on all four tasks.

M Influence of Mask IoU on Latent Action Quality

In practice, supervisory masks come from an off-the-shelf segmentation model whose boundaries are never pixel-perfect. To quantify how mask quality affects latent action alignment, we perturb ground-truth masks via morphological dilation (expansion) and erosion (shrinkage) with structuring element radii of 0 to 3 pixels. This drops the mask mIoU from 1.0 down to approximately 0.4–0.6

depending on the task and perturbation direction. We retrain MaskLAM@GT from scratch with each perturbed mask set and report the resulting NMSE on the DMW benchmark. We also report baselines as dashed horizontal reference lines to contextualize the absolute error level.

We do not separately perturb MaskLAM@SAM because SAM 2.1 masks already serve as the "imperfect mask" condition: the comparison between MaskLAM@GT (radius 0) and MaskLAM@SAM in Table 1 shows they are within noise, establishing that real segmentation errors from a foundation model are already tolerable (that too on datasets that do not exactly represent the kind of real-world data those models were trained on). The morphological perturbation experiment isolates a controlled, interpretable axis of degradation beyond what SAM produces.

Dilation vs. erosion. Mask expansion (dilation) includes a growing border of background pixels in the supervisory target. This introduces a small amount of exogenous signal into the loss, but the impact on NMSE is mild. Mask shrinkage (erosion) removes agent boundary pixels, reducing the informative area without adding exogenous noise; here the effect is similarly benign. Across all 10 DMW tasks and both perturbation directions, MaskLAM@GT remains below all label-free baselines at every radius tested, confirming that pixel-perfect segmentation is not required. Mask extraction methods with systematic boundary bias (e.g., coarser bounding-box prompts that over-segment by 1–2 pixels) are therefore safe to use with MaskLAM.

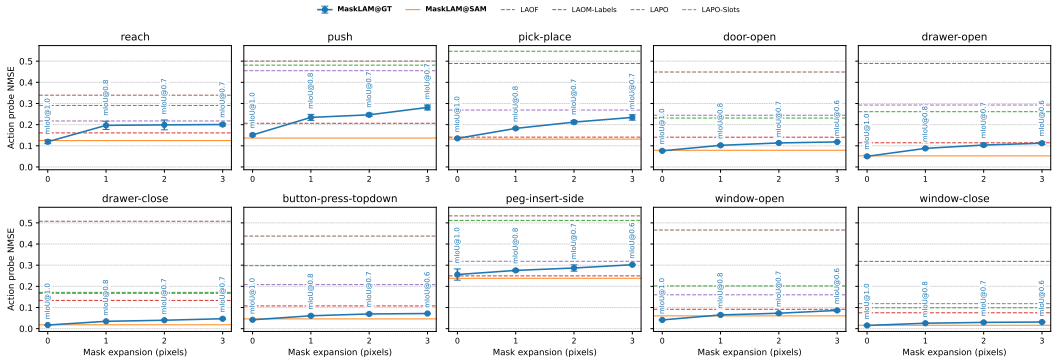

Figure 26: NMSE as a function of morphological dilation radius (0–3 pixels) applied to ground-truth masks before MaskLAM@GT training on the DMW. Each subplot is one task; the x-axis is the dilation radius in pixels (0 = unperturbed ground-truth mask); the y-axis is NMSE. Dashed horizontal lines show baselines. MaskLAM@GT remains below all label-free baselines across the entire perturbation range, indicating that including a thin border of background pixels in the loss does not meaningfully degrade latent action quality.

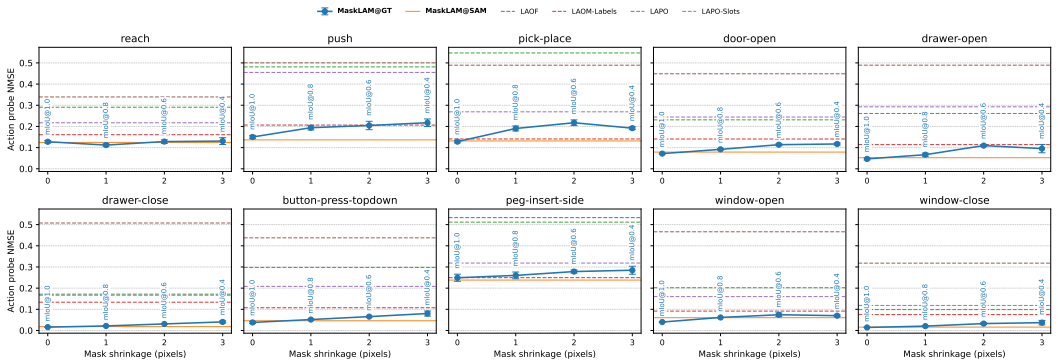

Figure 27: NMSE as a function of morphological erosion radius (0–3 pixels) applied to ground-truth masks before MaskLAM@GT training on the DMW. Layout matches Figure 26. Erosion removes agent boundary pixels, reducing the supervisory area without introducing exogenous signal. MaskLAM@GT degrades gracefully and stays below all label-free baselines even at radius 3, where mIoU drops below 0.5 on several tasks.

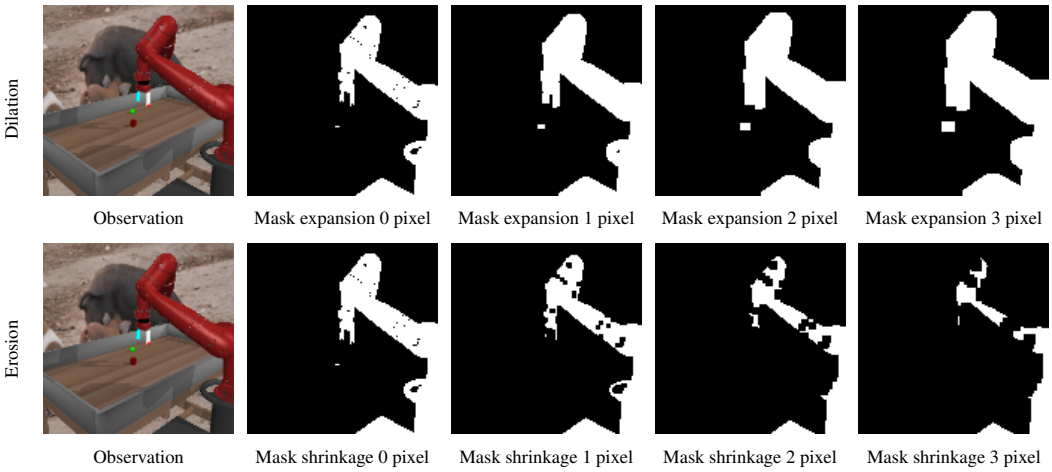

Figure 28: Qualitative visualization of morphological mask perturbations on a representative DMW frame (window-open task, distractor setting). Top row: progressive dilation of the ground-truth mask by 0–3 pixels, which expands the white supervisory region into the background. Bottom row: progressive erosion by 0–3 pixels, which shrinks the supervisory region inward from the boundary. Even at radius 3 the mask retains the coarse agent silhouette, explaining the graceful degradation observed in Figures 26 and 27.

N Extended Ablation Results

Per-environment breakdown of the aggregated ablation in Figure 8; variant definitions in Section 6.1. The findings of Section 6.1 hold at the per-task level on every DCS task (Figure 31) and every DMW task (Figure 30), as well as when aggregated separately by benchmark suite (Figure 29).

Loss masking is the principal contributor on every task. Removing the masked loss more than doubles NMSE on nearly all DCS and DMW tasks.

The mask channel helps on DCS but slightly hurts on DMW. With loss masking active, removing the mask channel increases NMSE on every DCS task, with the largest effect on cheetah-run and hopper-hop. We attribute the per-task variation to the thin limb geometry of these agents, where an explicit spatial prior helps the model isolate the agent from the dynamic background. On DMW, the channel slightly hurts performance instead, as the larger and more compact arm silhouettes already provide a sufficient prior. Without loss masking, the channel makes no meaningful difference on any task: it is useful only when the loss already restricts gradients to the masked region.

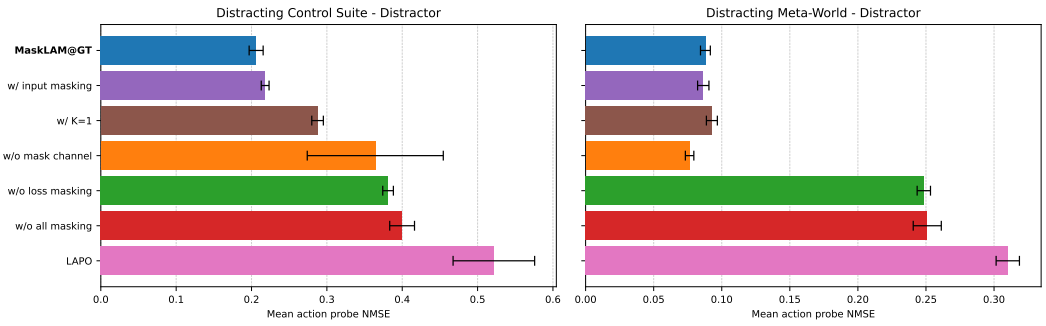

Figure 29: Ablation of MaskLAM components aggregated per benchmark suite (DCS left, DMW right). Loss masking is the principal contributor on both suites; the mask channel helps on DCS but slightly hurts on DMW; multi-step IDM adds a smaller orthogonal gain.

Multi-step IDM contributes a roughly additive gain. Setting $k{=}1$ degrades NMSE by a similar margin whether or not masking is present, consistent with the decorrelation argument of Lamb et al.

[15] and confirming that multi-step prediction complements rather than substitutes for the spatial intervention.

Input masking matches loss masking on every task. *w/ input masking* is within noise of MaskLAM@GT on every DCS and DMW task, mirroring the aggregate result in Section 6.1. Both benchmarks feature self-driven agents whose motion is generated entirely within M_t, so the interaction surface outside the mask carries no action-relevant signal beyond the agent pixels. Loss masking still preserves the IDM's access to the full scene, and Figure 32 confirms it attends to ground-contact pixels outside M_{t+1} in practice. In real video where the agent's motion depends on external contact (a rider on a scooter, a commuter on an escalator), that surface carries action-relevant signal that input masking would discard at the input.

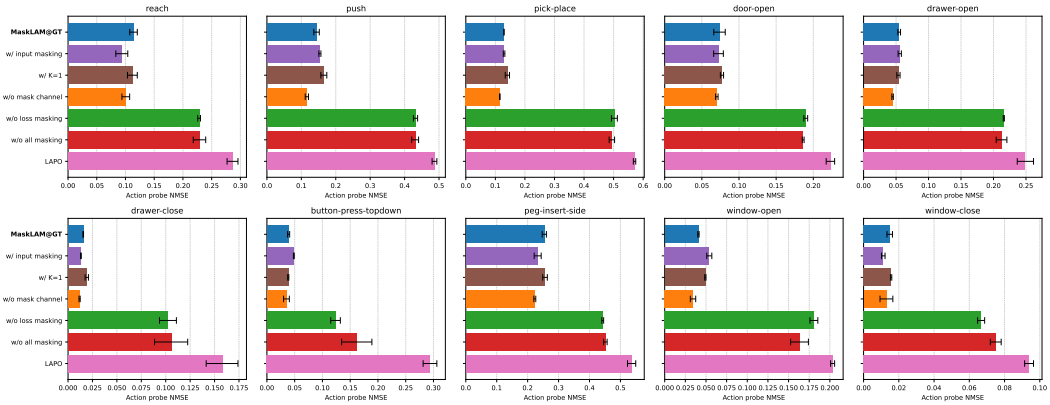

Figure 30: Per-task ablation on DMW (distractor). Loss masking is the principal contributor across all ten tasks; the mask channel slightly hurts performance, as the compact arm silhouettes already provide a sufficient spatial prior; multi-step IDM adds a smaller orthogonal gain.

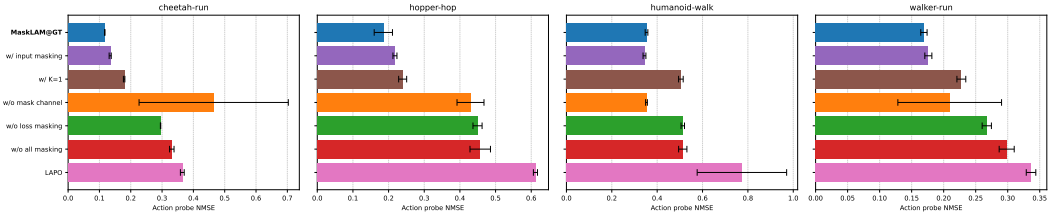

Figure 31: Per-task ablation on DCS (distractor). The hierarchy is consistent across cheetah-run, hopper-hop, humanoid-walk, and walker-run: loss masking is the principal contributor, followed by the mask channel, followed by multi-step IDM. The mask channel's contribution is largest on cheetah-run and hopper-hop, where the thin limb geometry makes an explicit spatial prior most useful.

O Extended Eigen-CAM Results

Eigen-CAM [19] produces a saliency map by taking the principal component of the activations of a chosen convolutional layer, highlighting the spatial regions that drive the network's prediction without requiring class labels or gradient signals. We apply it to the final convolutional block of the IDM and overlay the resulting heatmap on the input frame; warmer colors mark pixels that contribute most to the predicted latent action z_t. Since the IDM is the only module that consumes observations to produce z_t, its saliency directly exposes which pixels shape the latent action.

Figure 32 extends the single-frame comparison of Figure 5 to ten consecutive frames of a cheetah-run rollout in the distracting setting. LAPO's saliency drifts across the distractor background throughout the trajectory and rarely overlaps with the agent, confirming that without the masked loss the IDM grounds z_t in exogenous variation. MaskLAM's saliency stays locked onto the cheetah body and

onto the ground-plane region directly under its feet across all ten frames, even though the supervisory mask covers only the cheetah itself (see Figure 34). The ground-plane response is informative: it is endogenous context, the surface the agent interacts with to produce its motion, and the IDM has learned to attend to it without ever being told to.

This is the behavior we designed for in Section 4. The endogenous context varies from video to video and from frame to frame: the floor under a cheetah, a cup approached by a gripper, a door touched by a hand, a tool brought into contact with a workpiece. Specifying these regions a priori for every observation would require dense, task-aware annotation that defeats the purpose of label-free latent action learning, and any conservative mask drawn at input time would either erase useful interaction surfaces or admit distractors back in. Loss masking is therefore the only viable option: the IDM still sees the full observation and is free to attend wherever reconstruction demands, while z_t is shaped only by reconstruction errors inside M_{t+1}. The boundary of endogenous context emerges from data rather than from a hand-crafted mask.

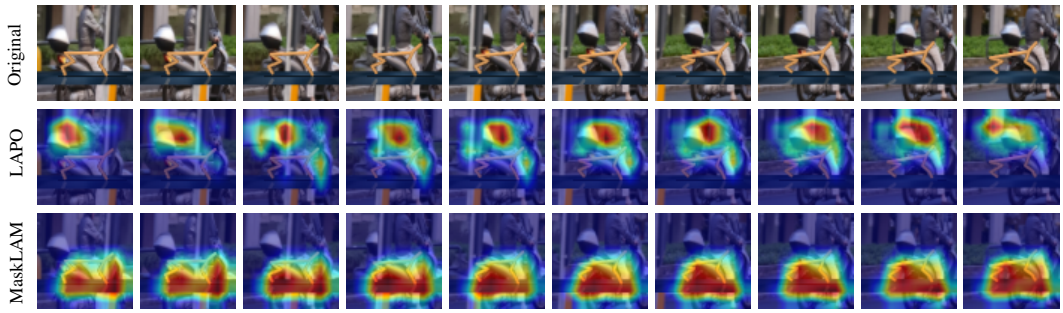

Figure 32: Eigen-CAM saliency on ten consecutive frames of a cheetah-run rollout on DCS (distractor). Warmer colors indicate higher influence on the predicted latent action. Top: Original frames. Middle: LAPO drifts across the distractor background. Bottom: MaskLAM stays on the cheetah and the ground-plane region under its feet, despite the supervisory mask covering only the cheetah.

P Extended UMAP Results

Figure 33 extends the single-dimension comparison of Figure 4 to all six action coordinates of cheetah-run on DCS (distractor). Each row corresponds to one method and each column to the ground-truth action dimension used to color the projection; a well-aligned latent exhibits a smooth color gradient along the manifold for every column.

MaskLAM recovers a coherent gradient on every action dimension. MaskLAM's manifold shows a clean red-to-blue transition for all six coordinates, with the gradient direction rotating from column to column as expected from a latent that is linearly aligned with the action vector. The per-dimension consistency confirms that the single-dimension view in the main text is representative rather than cherry-picked.

LAOM-Labels matches the structure using action supervision. LAOM-Labels also forms a connected manifold with visible color separation on every dimension, consistent with its use of ground-truth actions as an auxiliary supervisory signal during pre-training (Section 5). MaskLAM reaches comparable per-dimension structure without consuming any action labels at the latent action learning stage.

LAPO and LAOF collapse into entangled fragments with no action structure. LAPO's projection breaks into thin disjoint strands with no visible gradient on any dimension, and LAOF mirrors this pattern at finer fragmentation. Under distractors, the unmasked reconstruction loss steers both latents toward exogenous variance instead of agent-controlled dynamics (Section 6), and the per-dimension view shows this happens uniformly across the action vector rather than on isolated coordinates.

LAPO-Slots recovers partial structure on a subset of dimensions. LAPO-Slots produces a single connected manifold with locally visible red-blue regions on some coordinates, but the gradient is non-monotonic or washes out on the remaining ones. Object-centric decomposition isolates the agent

at the input, yet the unmasked reconstruction loss still allocates the latent's capacity unevenly across action coordinates.

The per-dimension picture matches the aggregate probe error reported in Table 1: only MaskLAM and LAOM-Labels recover a latent action space that is linearly decodable to the full action vector, and MaskLAM does so without any ground-truth action supervision during pre-training.

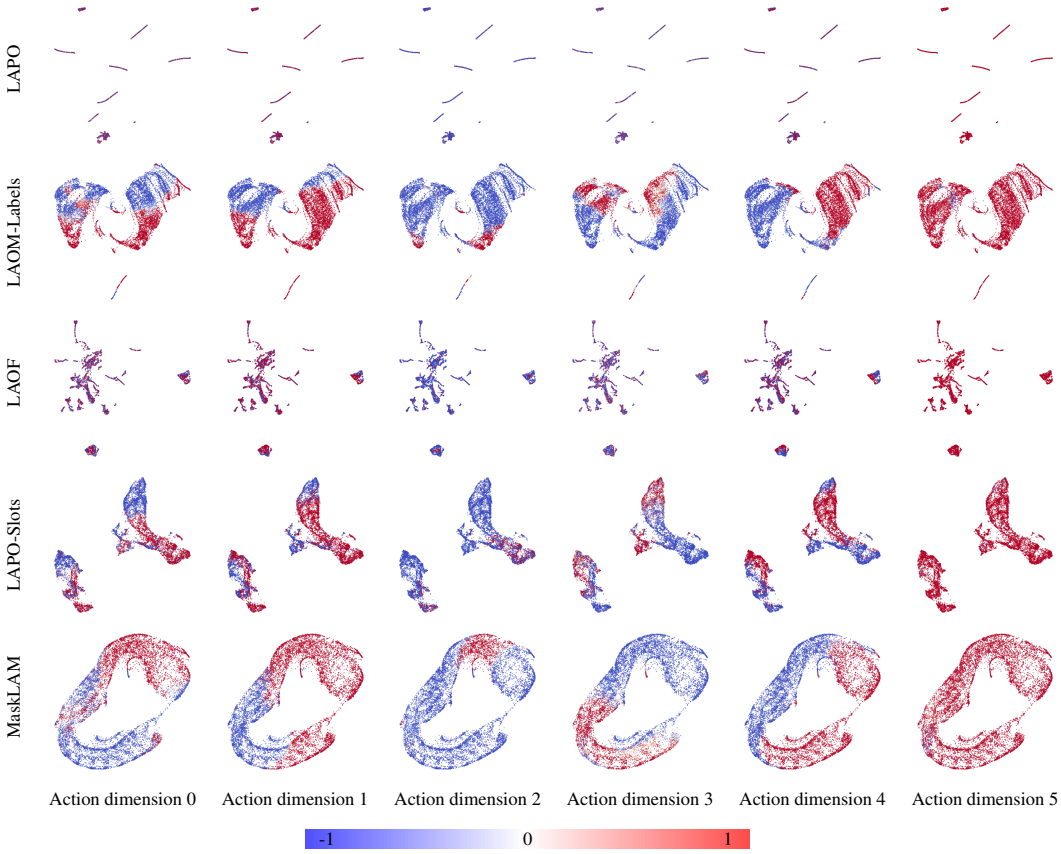

Figure 33: UMAP projections of learned latent actions on the DCS (distractor) dataset. Rows show LAPO, LAOM-Labels, LAOF, LAPO-Slots, and MaskLAM, respectively; columns show action dimensions 0–5.

Q Dataset Generation

We construct one offline dataset per benchmark and reuse it for every method in this work. To isolate the effect of visual distractors, we generate *synchronized* pairs of distractor-free and distracting trajectories: the underlying agent behavior and environment state are identical in both settings, so any gap in latent action quality or downstream return must come from how a method handles the visual distractors rather than from a different action distribution.

Expert policies. On DCS we reuse the released expert checkpoints of LAOM [20] rather than retraining our own. Briefly, those checkpoints are PPO [25] from CleanRL [13] for cheetah-run, walker-run, and hopper-hop, and SAC [10] from Stable-Baselines3 [22] for humanoid-walk, all trained on proprioceptive states with default hyperparameters; we refer the reader to [20] for the full training schedule. On DMW we use the released single-task SAC experts of Meta-World [34], one per task, trained with the Garage configuration of 500 epochs $\times$ 40 epoch cycles $\times$ 500 gradient steps per cycle, corresponding to roughly 10M environment steps per task; full hyperparameters are reported in [34]. Across both benchmarks the experts act on proprioceptive states, not on rendered images, so visual distractors cannot influence their decisions and the resulting action sequences are deterministic functions of the seed and initial state.

Trajectory collection. We roll out each expert checkpoint and render every state twice: once with a clean background and once with the distractor configuration of Section 5, sharing the camera pose, agent state, and time step across both renders. Because the expert ignores pixels, the two renders differ only in the observation channel and not in the recorded action or the next state, yielding the synchronized distractor-free / distracting pair described above. The collected datasets contain 9M training and 1M held-out transitions on DCS, and 1M training and 100k held-out transitions on DMW (Section 5). For each transition we store the rendered observation, the ground-truth simulator mask, the proprioceptive state, and the executed action.

Distractor videos. The dynamic backgrounds are sampled from DAVIS [21]. We assign DAVIS clips to our train and test splits disjointly: training trajectories sample backgrounds only from the DAVIS train split, and held-out evaluation trajectories sample backgrounds only from the DAVIS test split. This is intentionally stricter than the standard protocol in LAOM, which reuses the DAVIS clips seen during training in its in-distribution evaluation and reports a separate, harder "OOD" number on held-out clips. By construction, every evaluation in this paper corresponds to that OOD setting, so our reported metrics measure generalization to unseen distractor videos rather than memorization of the training backgrounds.

Masks. Ground-truth segmentation masks are read directly from the simulator at render time, so they are tightly aligned with the rendered observation and require no additional processing. SAM 2.1 masks [23] are produced in a separate post-processing pass over the released observations using the prompt-based extraction protocol detailed in Section R. All datasets, including ground-truth and SAM masks, will be released upon acceptance.

R Mask Generation

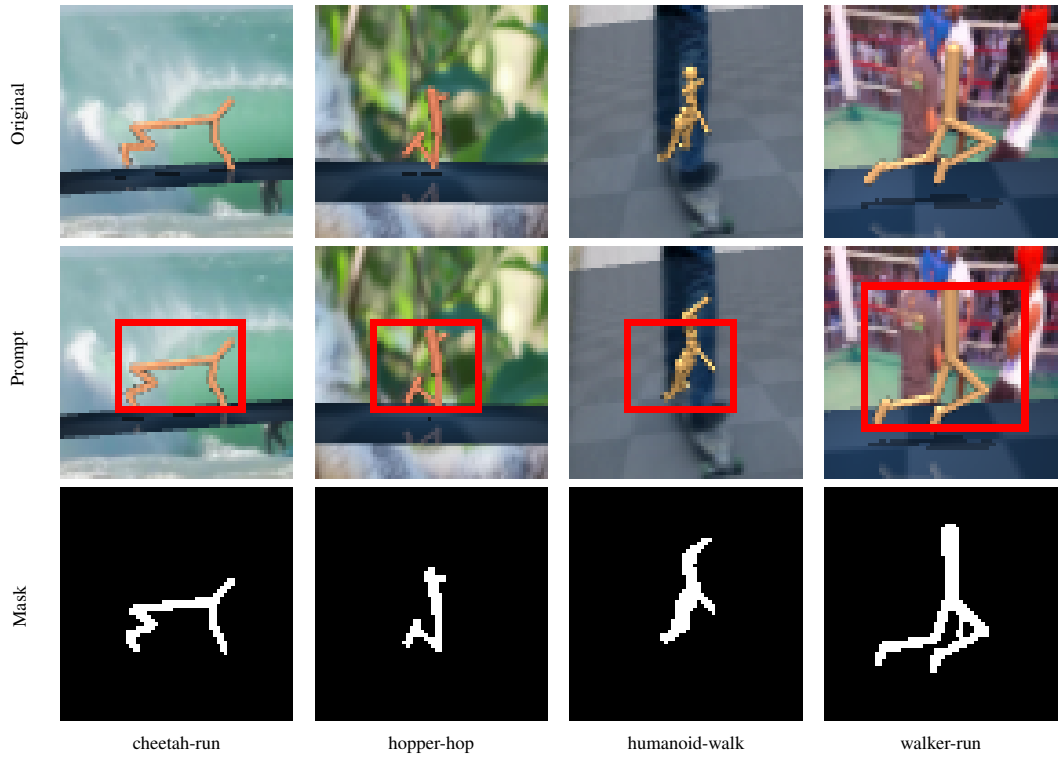

Figure 34: SAM 2.1 mask generation on DCS. Each column shows one task (cheetah-run, hopper-hop, humanoid-walk, walker-run); rows show, top to bottom, the rendered observation, the same observation with the fixed per-task bounding-box prompt overlaid, and the segmentation mask propagated by the SAM 2.1 video tracker from that single first-frame prompt. A single coarse box per task is sufficient to recover a tight agent mask across all four environments despite dynamic backgrounds, agent color randomization, and camera shake.

We pre-compute SAM segmentation masks for every trajectory in both benchmarks using the SAM 2.1-hiera-tiny video model [23]. To avoid per-frame annotation, we exploit the deterministic initialization of each environment: the agent always starts at a known location in the camera frame, so a single coarse, fixed-size bounding box on the first frame of a trajectory is sufficient to identify it. We feed this bounding box as the initial prompt to SAM 2.1 and let the video tracker propagate the mask through the remaining frames. The same procedure is applied to DCS and DMW; per-task bounding boxes are reused across all trajectories of that task. Figure 34 illustrates the prompt and the resulting segmentation for the four DCS environments.

At the reported throughput of 91.5 FPS for the 38.9M-parameter tiny variant on an NVIDIA A100 [23], annotating the full corpus takes approximately 309 GPU-hours on a single A100. SAM masks are stored alongside the simulator ground-truth masks in the released datasets so that any method can be evaluated under either mask source without re-running segmentation.